

Competitive Gamification in Digital Consumption: Evidence from TikTok

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July 10, 2026

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Abstract

We study the gifting behavior of consumers and leaderboard design in the context of livestreaming. Using real-time data from TikTok, we provide causal evidence for two distinct gifting motives: the competitive motive and the appreciation motive. The appreciation motive reflects the tendency to gift in response to popular content. The competitive motive arises from TikTok’s gifting leaderboard. The leaderboard ranks consumers by cumulative gifting in real time and awards badges to the top three contributors, creating incentives to compete for recognition and social utility. We find that the strength of the competitive motive depends on the consumer’s relative position: when a consumer’s cumulative gifting amount is far from that of immediate competitors, the return from competing diminishes, reducing the competitive motive. To evaluate the economic value of the leaderboard and guide the design of the leaderboard, we build and estimate a continuous-time dynamic game model of consumers’ gifting behavior. Our results show that, for an average session, the leaderboard-induced competitive motive accounts for 43% of the total revenue. We also conduct leaderboard design to manage the competitive intensity and optimize the platform revenue. We find that revising the scoring system to weigh recent gifting activity more heavily increases revenue by 19.7%. Our findings underscore the critical role of leaderboards in driving revenue on digital platforms and the importance of leaderboard design in optimizing platform revenue.

Keywords: Gamification, competition, livestreaming, leaderboard, moment-to-moment data, high-frequency identification, continuous-time dynamic game

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1 Introduction

Leaderboards are widely used on digital platforms to organize and publicly display users’ relative performance. For example, the livestreaming platform TikTok LIVE uses leaderboards to rank viewers in real time based on their cumulative gifting contribution within a live session, with virtual badges awarded to the top contributors. The fitness platform Peloton uses leaderboards to rank users during workout sessions based on calories burned, while the education platform Kahoot uses leaderboards to rank participants during quiz sessions by learning points. In both Peloton and Kahoot, a small number of top performers are prominently visible on the leaderboard. Across these platforms, leaderboards share two important common features. First, consumers see their effort ranked and displayed in real time during the session, based on activity-specific performance measures (e.g., gifting, calorie expenditure, or quiz performance). Second, consumers gain social utility from being ranked highly during the session, through increased visibility or badges awarded to top-ranked users. By embedding leaderboards into the consumer journey, platforms aim to foster social comparison and competitive instincts ([Festinger, 1954](#); [Liu, Santhanam, and Webster, 2017](#)), thereby increasing costly user engagement.

To be effective, leaderboards must be carefully designed. Prior research has studied various design dimensions, including award structures, competition size, and feedback frequency. A particularly important yet relatively overlooked design element is the scoring system, which maps participants’ histories of effort into a single performance metric that determines their rank. In many digital platforms, leaderboards rely on cumulative scoring, under which past effort permanently accumulates over the course of a session. A common concern with this design is stagnation, defined as low turnover among users occupying top-ranked positions.¹ For instance, in livestreaming sessions, viewers who contribute heavily early on may establish large score gaps, weakening incentives for both trailing users who find it costly to catch up and leading users who face little need to defend their positions. Importantly, however, stagnation is not unambiguously undesirable. Cumulative scoring also strengthens incentives for early effort, as participants rationally anticipate that initial contributions can translate into sustained competitive advantage. These opposing forces imply a central design problem: whether and how leaderboards should be redesigned to manage stagnation in order to improve overall user engagement and platform performance.

The goal of this study is to identify the competitive motive induced by the leaderboard and design alternative scoring systems to manage stagnation and sustain competition. More specifically, this paper addresses the following research question in the context of TikTok LIVE: (1) Does the leaderboard lead to competition among viewers? To what extent does stagnation occur? (2) What is the economic value of leaderboards in contributing to platform

¹For example: [Your Leaderboards Probably Won’t Work: Get it Right or Skip Them](#)

revenue? (3) How can leaderboards be better designed to manage stagnation and sustain competition, and how much revenue is going to improve as a result?

To answer these questions, we combine reduced-form analysis with a structural modeling approach. Using high-frequency data on viewers’ gifting behavior, we first document key empirical patterns that shed light on viewers’ gifting motives. We then develop and estimate a continuous-time dynamic game in which viewers within a livestreaming session strategically decide whether and how much to gift, taking into account their beliefs about the actions of other viewers. Finally, we use counterfactual simulations from the estimated model to quantify the revenue contribution of leaderboards and to evaluate how alternative scoring rules reshape gifting dynamics and session-level revenue.

Our work has two main contributions. The first contribution of this paper is to shift the focus from evaluating the effects of leaderboards to studying their design. We model the dynamic competition induced by leaderboards and evaluate alternative scoring systems as design tools to manage stagnation and sustain competition. In contrast, most existing studies treat leaderboards as a fixed feature and focus on documenting their impact on outcomes such as fitness (Hydari, Adjerid, and Striegel, 2023; Bojd et al., 2022), education (Li et al., 2024), and sales (Ho, Liu, and Wang, 2023). The second contribution is that our paper adds to the literature on dynamic contest design, in which rewards are distributed during the competition rather than at the end. In contrast, past literature on dynamic contest design has focused on rewards received only after the contest is over,² including labor competition (Tong and Leung, 2002) and crowd sourcing competition (Lemus and Marshall, 2021; Jiang, Huang, and Beil, 2022), etc. Therefore, prior learning on contest design provides little guidance for our study context, where the reward is the value of being in top positions on the leaderboard during the session, potentially because of the social recognition it brings (Festinger, 1954; Liu, Santhanam, and Webster, 2017).

En route to developing our structural model, we discuss why viewers are willing to gift content creators voluntarily, and document the stagnation when viewers compete through gifting on TikTok LIVE. First, we show that viewers value higher leaderboard ranks and are willing to gift to climb the rankings, which we term the competitive motive. To uncover viewers’ preferences for the rankings, we use a revealed-preference approach to examine how their gifting behavior changes when their rankings are exogenously demoted on the leaderboard. The underlying logic of this approach is that if a viewer increases her gifting amount after being demoted from rank k to rank $k + 1$, she must strictly prefer rank k to rank $k + 1$, since she is willing to incur an additional monetary cost to regain her old ranking. However, the challenge is to find such exogenous variations in the ranking, especially when the ranking is inherently endogenous. To address endogeneity, we exploit an institutional feature of TikTok LIVE: the leaderboard ranks viewers only for those who are present in the

²See Dechenaux, Kovenock, and Sheremeta (2015) for a review of contest design.

session. As a result, if a high-contributing viewer temporarily exits and later reenters, her return will bump down the rankings of all other viewers with lower scores. In the analysis, we examine the gifting behavior of those viewers whose rankings are involuntarily bumped down by the re-entrants and find that they increase gifting to regain their prior rank, consistent with a preference for higher ranks.

Heterogeneous analysis of re-entry events provides direct evidence for stagnation. It reveals a negative relationship between competitive motive and score gaps between ranks: bumped viewers are less likely to increase gifting when their scores are further away from the re-entrants'. Based on this finding, we develop two measures of stagnation to examine its dynamics within the session. First, for each $k \in \{1, 2, 3\}$, we compute the score gap between rank k and rank $k + 1$. Larger gaps indicate greater difficulty for lower-ranked viewers to overtake those above them. Second, we compute the turnover rate of the top-three positions—i.e., the frequency with which the identity of the top-three players changes due to gifting competition. Lower turnover reflects greater stagnation, as it indicates that top positions are effectively “locked in.” By plotting the stagnation measures against the session progression, we find that stagnation intensifies over the course of a session. Score gaps between adjacent ranks grow substantially, while the turnover rate of the top three declines over time.

Lastly, we find evidence of an additional motive: viewers may gift simply out of appreciation for the content creator. Specifically, they tend to gift more when content quality is higher.³ This pattern suggests that viewers derive utility from gifting as an expression of appreciation for the high-quality content.

Building on these reduced-form findings, we develop a continuous-time dynamic game that embeds the empirically documented motives into a unified framework. The model is therefore not imposed a priori, but is grounded in model-free evidence on how viewers respond to leaderboard ranks, score gaps, and content quality. We develop a novel continuous-time dynamic game model to capture viewers' gifting behavior. In our model, each viewer can endogenously decide whether they want to exit at any moment during the session. If they choose to stay in the session, they can decide how much to gift. They have the option to stay in the session without gifting anything. The leaderboard updates immediately following each action, removing the score if the viewer exits, refreshing with an increased score if she gifts, or remaining unchanged if she does nothing. We model play using continuous time rather than a discrete time game for two main reasons. First, the continuous-time model can better capture viewers' decision processes: the session evolves continuously, and viewers can make exit and gifting decisions at any time during the livestreams. Second, the continuous-time model can drastically reduce the computation burden for dynamic game problems, as

³We focus on viewers ranked below five who have little to no competitive motives, given they are not near obtaining the badges, to provide a clean identification of the appreciation motive.

players move sequentially in the continuous-time model rather than simultaneously in the traditional discrete-time model.⁴

Viewers’ decisions are modeled as the outcome of a dynamic optimization problem, in which they compete with each other over time in each session. The competitive motive induces a long-term tradeoff in gifting. This is because viewers are modeled as forward-looking agents and they weigh the immediate disutility of spending against the effect of their score on subsequent competition. For example, gifting a lot may help a viewer obtain top-three positions on the leaderboard, increasing her subsequent utility from holding the badges.⁵ In contrast, gifting out of appreciation only induces a short-term tradeoff in gifting: at the moment of gifting, viewers trade off the disutility of spending against the immediate utility of expressing appreciation to the content creator. Viewers’ gifting is a joint outcome of these two types of motives. Moreover, our model flexibly accommodates a wide range of events that shape the session’s evolving states, including entry, exit, re-entry, gifting, and fluctuation in content quality.

Our estimation results quantify three economically meaningful sources of utility in livestreaming: competitive motive in gifting, appreciation motive in gifting, and intrinsic consumption value from watching. First, viewers place substantial value on leaderboard positions. On average, each minute of staying in the rank-1 position is worth 0.660 dollars, while staying in the rank-2 or rank-3 positions is worth 0.203 dollars per minute, indicating a sharp drop-off in value beyond the top rank. Second, the utility from showing appreciation through gifting increases strongly with content quality. A one-standard-deviation improvement in content quality raises the utility of appreciation-motivated gifting by approximately 50% relative to its baseline level where the content quality is at its median. Finally, viewers derive sizable consumption value from watching the livestream itself: at median content quality, watching generates 0.116 dollars per minute, and a further one-standard-deviation increase in quality raises this value by an additional 0.089 dollars per minute.

After estimating the structural model, we use it to perform counterfactual simulations which address the paper’s main managerial questions: First, what is the value of the leaderboard in terms of platform revenue? Second, can the platform better manage stagnation and improve revenue through alternative scoring rules? To answer the first question, we remove the competitive motive induced by the leaderboard by setting the rank-based utility to zero and examining how much revenue drops. We find that the total revenue for an average session is \$40, and the competitive motive accounts for 43% of total revenue. This finding

⁴In continuous-time models, the probability that multiple agents make decisions at exactly the same moment is zero.

⁵In our model, the competition is driven by the utility of badges on the leaderboard, which are backed by the institutional fact that only top-three contributors are awarded with badges, and our reduced-form results also indicate viewers care about getting into top-three positions, i.e., rank 1 $\succ$ rank 2 $\sim$ rank 3 $\succ$ rank 4 $\sim$ rank 5 and beyond.

highlights the central role of leaderboards in generating revenue on TikTok LIVE. We further conduct heterogeneous analysis by dividing the sessions into long sessions and short sessions, and find that leaderboards play a less important role in generating revenue for long sessions than short sessions, i.e., the percentage of revenue from the leaderboard and the revenue per minute generated by the leaderboard are both smaller in long sessions. The reason is that long sessions suffer from a stagnation problem more severely; i.e., leaderboards become stagnant when top contributors are far ahead on the leaderboard, making it seem futile to exert additional effort in gifting.

The second counterfactual investigates whether and how alternative scoring rules can be used to manage stagnation in gifting competition. Importantly, leaderboard design is a non-trivial problem because managing stagnation involves a fundamental trade-off between early and late gifting incentives. On the one hand, stagnation discourages gifting by lower-ranked viewers, who perceive little chance of catching up or little threat to their current rank. On the other hand, because viewers are forward-looking, stagnation can strengthen incentives to exert effort early in the session: early contributions can persist over time and translate into sustained competitive advantage. As a result, reducing stagnation may improve participation later in the session while simultaneously weakening incentives to gift early.

To study how this trade-off can be managed through leaderboard design, we consider a discounted score scheme that gradually discounts past gifting contributions over time, placing greater weight on recent activity and less weight on earlier contributions when ranking viewers. Operationally, the platform discounts each viewer’s accumulated score every 30 minutes by multiplying it by a discount factor β .

Counterfactual simulations show that discounted scoring improves platform revenue only for an intermediate range of discount factors. Revenue is maximized at $\beta = 0.55$, which increases total session revenue by 19.7% relative to the baseline. When β is large and discounting is weak, early gifting contributions remain highly persistent, allowing score gaps between viewers to widen over time and leading to stagnation in later stages of the session. Conversely, when β is too small and discounting is aggressive, forward-looking viewers anticipate that early contributions will rapidly lose their influence on future rankings, reducing their incentive to gift in the first place. The optimal discount factor balances these opposing forces. Heterogeneity analysis further shows that longer sessions benefit more from discounted scoring than shorter sessions, as prolonged competition exacerbates stagnation and calls for more aggressive discounting (i.e., a smaller β).

Our counterfactual results will have direct implications for managers of livestreaming platforms, which is a significant market on its own: it was valued at \$88 billion in 2024 and is projected to grow at 24% annually, reaching \$600 billion by 2033 (Imarc, 2025). Importantly, voluntary payments constitute the primary revenue source for leading livestreaming platforms such as Twitch (70%), TikTok LIVE (>99%), Bigo Live (>80%), and Kuaishou

(90%). These payments are later split between the platform and content creators, meaning that the incentives of streamers align with those of the platform. The results highlight the economic importance of leaderboard design in shaping gifting behavior and revenue outcomes. In particular, they suggest that discounted scoring can serve as an effective design lever for managing viewer competition and mitigating stagnation, thereby increasing revenue for both platforms and creators.

2 Related Literature

Our paper contributes to four strands of literature. The first strand is on competitive gamification and leaderboards. The second strand is on dynamic contest design. The third strand is on the emerging literature on the rapidly growing livestreaming industry. The last strand is on the use of moment-by-moment data.

I. Competitive Gamification & Leaderboards Our work contributes to the growing literature on leaderboards and competitive gamification. A large body of recent research studies the effects of leaderboards across a range of contexts, including health and fitness (Hydari, Adjerd, and Striegel, 2023; Bojd et al., 2022), sales (Ho, Liu, and Wang, 2023), and education (Santhanam, Liu, and Shen, 2016). These studies generally find that leaderboards increase engagement or performance on average, while generating substantial heterogeneity across users. However, this literature predominantly adopts reduced-form approaches that take leaderboard design as given, focusing on whether leaderboards work rather than how they should be designed.

A smaller set of papers employs structural approaches to study competitive responses induced by leaderboards and related gamification mechanisms. For example, Lemus and Marshall (2021) shows that providing information about contestants’ relative performance increases both the quantity and quality of submissions on a crowdsourcing platform, and Wang, Zhang, and Ho (2025) study how badge design affects content production on Stack Overflow. While these studies provide valuable insights into competitive behavior, they do not quantify the economic value of leaderboards for platforms, nor do they examine how leaderboard scoring rules shape the dynamics of competition over time.

Our paper advances this literature by shifting the focus from the presence of leaderboards to their design. We quantify the revenue impact of leaderboards by structurally modeling viewer competition in livestreaming, and we study how alternative scoring rules can be used as design tools to manage the dynamics of competition. In particular, we document and model stagnation, the gradual weakening of competitive incentives as cumulative score gaps widen over time, and show how alternative scoring designs can mitigate this force and sustain competition, with direct implications for platform revenue.

II. Dynamic Contest Design A large theoretical literature on contest design studies how

rank-based prizes can be used to provide incentives, typically assuming tangible rewards such as monetary prizes or job promotions that are awarded at the end of the competition (Lazear and Rosen, 1981; Green and Stokey, 1983; Nalebuff and Stiglitz, 1983). Building on these foundations, a broad experimental and empirical literature examines how alternative dynamic contest designs shape effort and performance across a range of offline settings, including labor markets, sports, and education (Tong and Leung, 2002; Ehrenberg and Bognanno, 1990; Abrevaya, 2002; Szymanski, 2003; Sunde, 2009; Genakos and Pagliero, 2012; Leuven, Oosterbeek, and Van der Klaauw, 2010; Fershtman and Gneezy, 2011). More recently, a small but growing literature studies dynamic competition on digital platforms using field data, most prominently in crowdsourcing environments where contestants compete for well-defined tasks and monetary rewards distributed only at the end of the contest (Liu et al., 2014; Lemus and Marshall, 2021; Chan, Chen, and Wu, 2023; Lemus and Marshall, 2024). While these studies provide important insights into effort dynamics, they continue to focus on contests in which rewards are realized only upon completion.

In contrast, a substantial share of competition on modern digital platforms operates under a fundamentally different structure, in which rewards take the form of social utility (e.g., rank visibility, recognition, and status) and are realized continuously throughout the competition. Despite the prevalence of such settings on social media platforms, there is limited research on them. Our paper fills this gap by studying leaderboard-based competition in livestreaming, developing a structural model of dynamic competition under continuous social rewards. By shifting the focus from end-of-contest prizes to real-time social rewards, our analysis highlights a class of design problems that existing contest models are not designed to address.

Our finding of stagnation speaks to a central concern in the dynamic contest literature: revealing performance information can dampen incentives by exposing performance gaps. When contestants perceive a sufficiently low likelihood of winning, lagging players may optimally reduce effort or exit the competition altogether (Aoyagi, 2010; Fershtman and Gneezy, 2011; Goltsman and Mukherjee, 2011; Liu et al., 2014; Klein and Schmutzler, 2017; Jiang, Huang, and Beil, 2022). A common insight in this literature is that providing partial feedback by reducing its frequency can mitigate discouragement and sustain aggregate effort.

These approaches, however, are not applicable in our setting, where rank-based utility arises precisely because participants can observe their relative standing in real time. Instead, our paper proposes discounted scoring as a mechanism to sustain competition by reshaping the evolution of performance gaps while preserving real-time rank observability.

III. The Livestreaming Literature Our paper contributes to the growing literature on the emerging phenomenon of live streaming. Most existing papers study the effectiveness of live-streaming promotion. Huang and Morozov (2025) examine the impact of video game live streaming on the demand for the broadcasted games. Gu, Zhang, and Kannan (2024) study

firm’s strategy in choosing influencers for its marketing campaign and shows that there is a negative interaction effect between big and small influencers. [Liu \(2023\)](#) uses reinforcement learning method to create dynamic coupon targeting strategies in live stream shopping. [Liu et al. \(2022\)](#) investigate the lead indicators of the success and survival of livestream shopping sellers.

There is a small but growing strand of literature studying the determinants of gifting behavior in live streams. Existing studies emphasize the impact of viewers’ emotional states ([Lin, Yao, and Chen, 2021](#)) or social image concerns ([Lu et al., 2021](#); [Yao, Lu, and Chen, 2024](#)) on gifting activities. Our paper differs from this literature by adopting a structural modeling approach to capture gifting as the joint outcome of spontaneous responses to content and dynamic competition among viewers, and by examining how alternative leaderboard designs shape gifting dynamics and platform revenue. This approach allows us to move beyond reduced-form evidence and provide structural guidance on the optimal design of leaderboard-based competition in live streaming.

IV. Moment-by-moment Data Our paper contributes to the literature on the use of moment-by-moment data. Enabled by granular data, high-frequency identification has been widely used in contexts where the study objects are notoriously endogenous, like the effect of TV advertising ([Lewis and Reiley, 2013](#); [Liaukonyte, Teixeira, and Wilbur, 2015](#); [Joo, Wilbur, and Zhu, 2016](#); [Joo et al., 2014](#); [Du, Xu, and Wilbur, 2019](#); [Liu and Hill, 2021](#); [He and Klein, 2023](#); [Fossen and Schweidel, 2017](#); [Tirunillai and Tellis, 2017](#)), and the effect of monetary policy ([Bernanke and Kuttner, 2005](#); [Nakamura and Steinsson, 2018](#)). In this paper, we leverage high-frequency identification method to examine the effect of ranking.

The use of continuous-time models dates back to the 1970s, with most of the work focusing on the relation between underlying continuous-time data-generating processes and approximations by a discrete-time model([Phillips \(1972\)](#), [Sims \(1971\)](#)). [Heckman and Singer \(1986\)](#) argue for the use of continuous-time models instead of discrete-time models, motivated by a better match with the fact that the underlying data-generating processes are continuous-time and the frequency of data collection may not match with the frequency of decision making.

Although discrete-time choice models are still the standard in the economic literature in structural work, recent developments in data science and econometric methods provide more opportunities for continuous-time research. With the digitization of data, it is increasingly efficient to collect/store real-time data, which allows us to study the real data-generating process instead of forcing agents’ decisions to be simultaneous. [Doraszelski and Judd \(2012\)](#), [Arcidiacono et al. \(2016\)](#), [Blevins \(2016\)](#) recently contributed to estimation methods to accommodate discrete-time data collection into continuous-time modeling. Moreover, they demonstrate that continuous-time modeling can be especially powerful in solving dynamic stochastic games. By accommodating sequential decision making, continuous time avoids

the curse of dimensionality in dynamic games and speeds up the computation by orders of magnitude.

In marketing, continuous-time models have been primarily used to study the timing of viewer purchases through hazard models, explaining the timing of the purchase or adoption of products (Naik, Raman, and Winer, 2005; Sinha and Chandrashekar, 1992). The large majority of continuous-time models have been estimated on product adoption or purchase data sets, with the notable exception of Deng and Mela (2018), who modeled the process of TV viewing, and Nevskaya and Albuquerque (2019), who model the process of online gaming usage. We add to this literature by developing a continuous-time stochastic game model to explain the competition among viewers on gifting decisions and conduct counterfactual design on competition mechanics.

3 Institutional Background

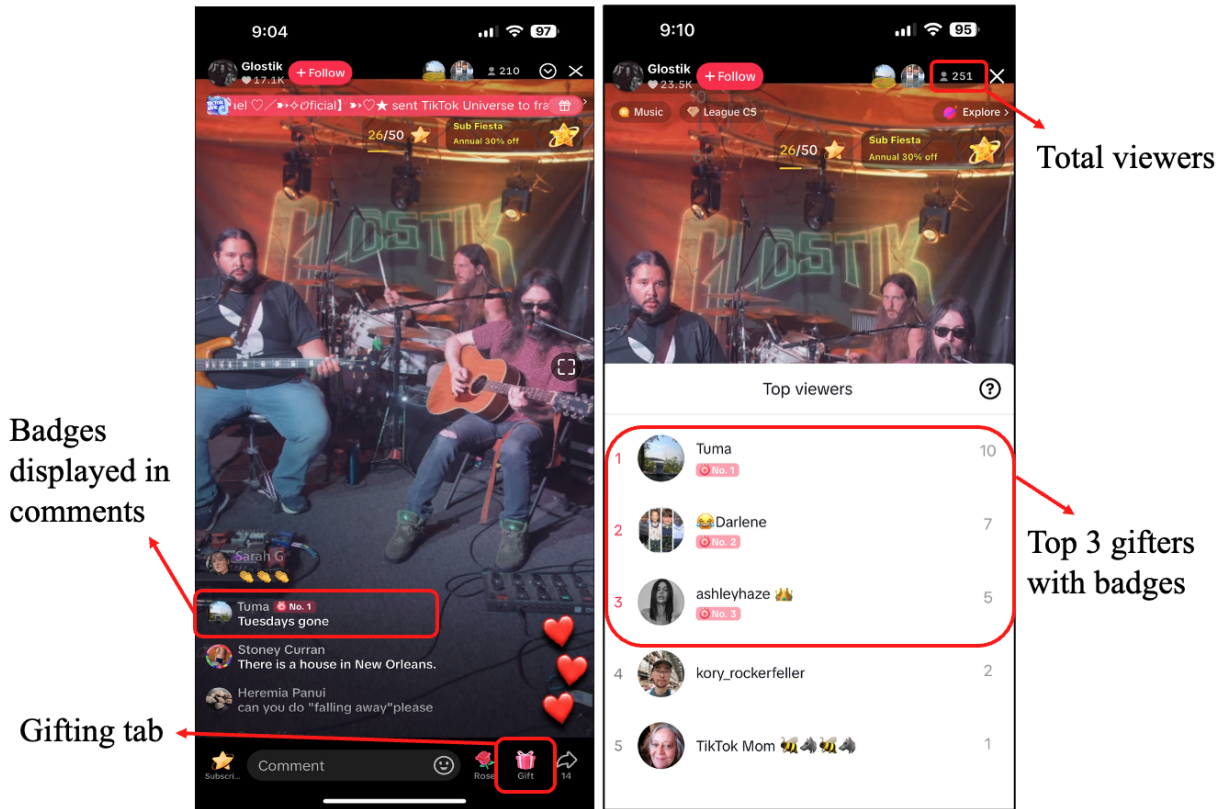
TikTok LIVE is the livestreaming platform in the TikTok app that lets creators broadcast video in real time to engage with their audience more personally and interactively. Viewers can join and leave the session anytime during a live stream, send likes and gifts, and post comments, all in real time. The dominant monetization of TikTok LIVE is the voluntary gifting system. Viewers can purchase virtual coins through the app to send virtual gifts to creators during a live stream. On average, one TikTok coin costs about 1.4 US cents. The total revenue is split between three parties: the App Store (30%), TikTok (35%), and the creator (35%). Creators can redeem the gifts they receive for dollars after commissioning the platforms.

The voluntary gifting system on TikTok LIVE continues to grow substantially, even as the platform holds the No.1 global ranking for in-app purchase revenue. According to the fourth-quarter digital marketing index report by Sensor Tower, viewers spent a staggering \$6 billion on virtual gifting in TikTok LIVE in 2024, up from \$4.4 billion in 2023, with a 36% year-over-year increase (SensorTower, 2024). At this pace, the revenue from virtual gifting is projected to surpass \$15 billion by 2027. While the platform offers other monetization tools, such as monthly subscriptions to creators, gifting accounts for the vast majority of its in-app purchases, underscoring its central role in driving platform revenue.⁶ In our data, we find that gifting constitutes over 99% of all revenue for the average creator on TikTok LIVE.

Figure 1 presents an example of the viewer interface on TikTok LIVE during a live music performance. The left panel shows the setup for a live music session. First, the content is free to access. The viewers can attend the session and watch the performance at no cost. Second, gifting is voluntary. They can click on the gifting tab at the bottom right of the screen to send out gift. Such gifting activities can happen anytime for multiple times during

⁶Subscribing to the creator unlocks special features, such as custom emojis and subscriber badges.

Figure 1: Viewer Interface of TikTok LIVE



the session. Third, viewers can tap the screen at any time to send likes, which appear as floating heart animations. Sending likes is free, anonymous, reflecting how much viewers like the content. Lastly, viewers can submit text comments via the comment box at the bottom of the screen. Commenting is an important way for viewers to interact with creators. In the example, viewers enter the names of songs they want to listen to, and creators monitor the comment section and pick one to perform. As a result, viewers' consumption utility is higher when their comments are answered by the creators.

The right panel shows the leaderboard setup. The competitive gamification on TikTok LIVE centers on a leaderboard and badges for top viewers. In the top-right corner, it displays the audience size. Tapping this number opens the detailed leaderboard, listing all viewers who have sent gifts (hereafter, *gifters*). The leaderboard ranks gifters by their cumulative gifting amounts in real time during the session, which we call the *score*.⁷ During a session, whenever a viewer sends a gift, the leaderboard automatically refreshes to track that viewer's cumulative gifting amounts up to that moment. At the end of each live stream, the leaderboard automatically resets and does not extend the record to future streams. The leaderboard displays up to top 99 gifters' identity, ranking, and cumulative gifting amount.

⁷The cumulative gifting amount is measured in the unit of virtual coins, and each virtual coin values 1.4 cents to purchase

The identities for the top 5 gifters are automatically visible, and you can get access to other gifters by scrolling down the leaderboard. The leaderboards’ rewards are in the form of badges. The top three gifters receive a visible ranking badge next to their names. For instance, viewer "Tuma" gifted 10 virtual coins and ranks No. 1, viewer "Darlene" gifted 7 and ranks No. 2, and viewer "Ashleyhaze" gifted 5 and ranks No. 3. There are two potential mechanisms for why viewers value the badges: one is social recognition (Lu et al., 2021). This is because creators will monitor the leaderboard in real time and acknowledge the generosity of top gifters, which is witnessed by all audience in the session. Second, badges can increase the likelihood that creators will interact with the badge owner through comments (Wohn, Freeman, and McLaughlin, 2018). Notably, comments from the top three gifters are accompanied by their ranking badges, increasing the likelihood that the creator will pay attention to them.

4 Data Description

It is particularly challenging to identify specific motives of viewers during a live stream due to the rapid changes in the livestreaming content. Therefore, we need to rely on moment-to-moment data and high-frequency identification.

Data Collection We developed a live-scraping software to collect livestreaming activities across many active creators and viewers. The software has two features: monitoring and recording. It first monitors if pre-selected creators turn on their live stream, then starts recording all the live stream activities in real-time. Due to the immense challenges of monitoring and recording individual-level activities in real time, we conduct data collection in two stages. In the first stage, we pre-select a sample of creators as follows. During the first week of May 2023, we scraped all the recommended creators on the TikTok LIVE front webpage, which displays a common set of recommended creators to all users rather than personalized recommendations. By restricting to creators with at least 5000 followers, our database includes 222 creators. In the second stage, for seven months from June to December 2023, we continuously monitored and recorded the live schedules of pre-selected creators and gathered information on the real-time activities of viewers attending their live sessions, resulting in a total of 32468 sessions with an average of 146 sessions per creator.

The collected data includes which creators are live, the topic of each session, the number of concurrent viewers at each moment, and all moment-to-moment viewer activities in the live stream, including gifting, liking, commenting, entry, and exit in real-time. For each activity instance, we record the viewer ID, live stream ID, creator ID, timestamp of activity, and activity information. The gifting activity information records the gifting value measured in the unit of virtual coins.⁸ For like activity, a viewer can press “like” at any point in time

⁸On TikTok LIVE, viewers have a balance of virtual coins and can charge the balance anytime at the

to signal a positive evaluation of the content at that moment. The liking activity records the number of likes a viewer sends. The commenting activity information records the message sent by the viewer. Entry and exit activities only include information on the timestamps.

Table 1: Summary Statistics of Major Activities

Variable	Mean	Std.Dev	Percentile				
			Min	25	50	75	Max
Panel A: Session level ($N = 32,468$)							
Session duration (minutes)	128	178	1	59	96	166	10445
Number of unique viewers	8600	42100	0	28	364	2726	1784518
Number of unique gifters	30	56	1	5	13	32	1159
Total gift value (coins)	7279	43940	1	76	767	5268	116538
Total likes	24265	51763	0	2252	7591	22954	1773973
Panel B: Gifter-session-minute level ($N = 31,705,888$)							
Gift value (coins)	12.10	402.23	0	0	0	0	419988
Gift indicator (1 = positive gift)	0.07	0.25	0	0	0	0	1
Gift value conditional on positive gift	154.30	1429.09	1	1	2	14	419988
Likes per minute	9.70	37.33	0	0	0	0	1947

Notes: This table provides summary statistics of our data in two panels. Panel A shows session-level sample statistics. It shows that the statistics on sessions are very skewed. Panel B shows the statistics when we restrict the moment-by-moment sample to include only individual-session-minute-level observations of gifters. This dataset in Panel B will be used in the following empirical analysis to analyze gifting activities. The summary statistics show that gifting behavior is still very sparse, even with this restriction.

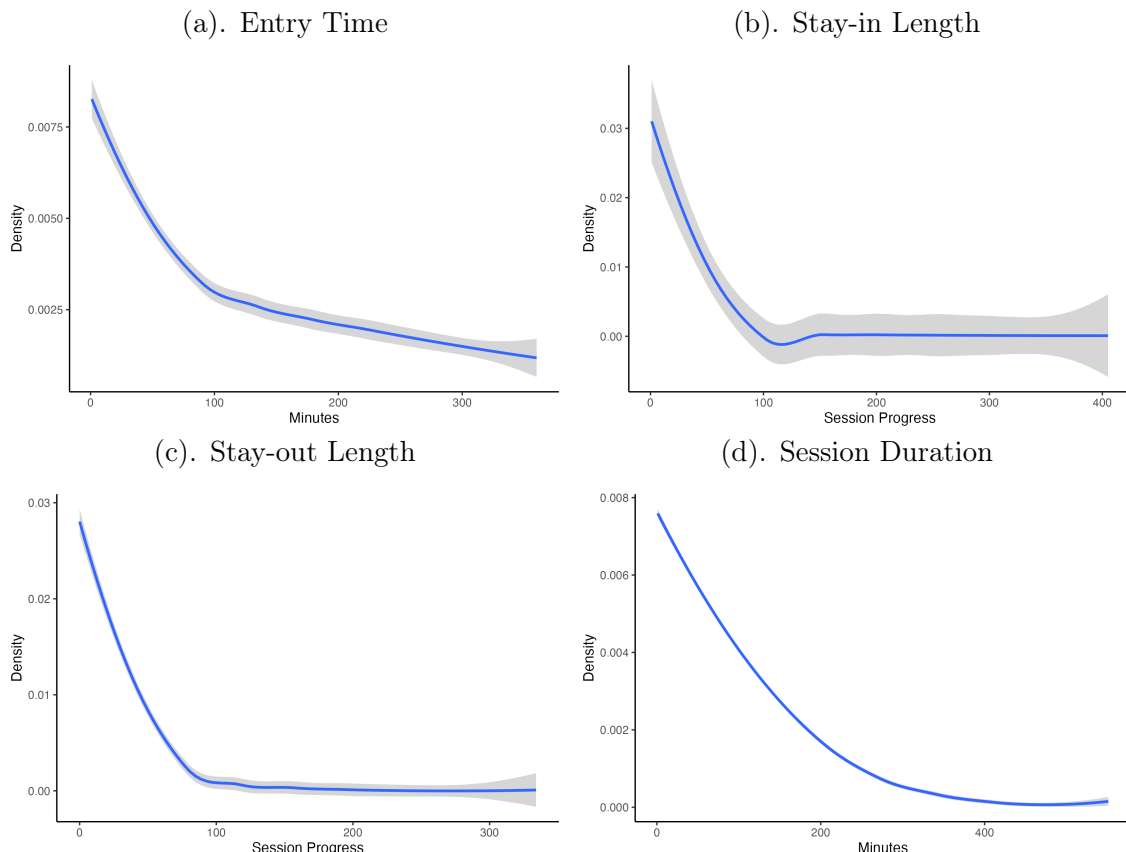
Summary Statistics of Major Activities Table 1 reports summary statistics for the main activities in our data, presented in two panels. Panel A shows session-level statistics. The distributions at the session level are highly skewed. The average session duration is 128 minutes, with a standard deviation of 178 minutes. An average session attracts 8,600 unique viewers, but only about 30 unique viewers send gifts to the creator. We refer to viewers who send positive gifts during a session as gifters for that session.

The distribution of total gift value (coins) across sessions is highly skewed, with a mean of 7,279 coins (approximately 100) and a median of 767 coins (approximately 11). Sessions also receive a large volume of likes: the average number of total likes per session is 24,265, which is about three times the median, reflecting substantial heterogeneity in session popularity and engagement.

Panel B reports summary statistics at the gifter-session-minute level, restricting the cost of \$0.014 per coin. Each gift’s price is listed in the unit of coin. Viewers with a positive balance can purchase the gifts during the live session. For instance, a rose is worth a virtual coin.

sample to minute-level observations of gifters. This dataset forms the basis of our empirical analysis. Despite this restriction, gifting activity remains highly sparse. Gifters make a positive gift in only about 7% of session minutes, as reflected by the mean of the gift indicator. Conditional on a positive gift, the distribution of gift value (coins) is extremely right-skewed: the mean gift value conditional on a positive gift is 154.30 coins with a standard deviation of 1,429.09, while the median conditional gift value is only 2 coins.

Figure 2: Distribution of Entry, Exit, Re-entry and Session Duration



Notes: This figure presents the distributions of entry time, stay-in length, stay-out length, and session duration over minutes, respectively. All four distributions follow an exponential distribution, which we use later in the structural modeling.

Timing of Gifter Participation In addition to the major activity measures, we observe the complete timing history of viewer participation. Specifically, we record each gifter’s entry and exit timestamps within a session, as well as the start and end times of each live-streaming session. These timing variables are central to both our empirical analysis and the specification of the structural model.

Using these timestamps, we construct four duration measures and jointly present their empirical distributions: (i) entry time, defined as the time elapsed between the start of a session and a gifter’s initial entry; (ii) stay-in length, defined as the duration between a gifter’s entry and subsequent exit from the session; (iii) stay-out length, defined as the duration between a gifter’s exit and her subsequent re-entry; and (iv) session length, defined

as the total duration of the live-streaming session.

Figure 2 presents the empirical distributions of the four duration measures described above, all measured in minutes. Panel (a) shows the distribution of gifter entry times relative to the start of the session. Entry activity is highest at the beginning of a session and gradually declines over time. Panel (b) shows the distribution of stay-in length. On average, gifters remain in a session for 44 minutes before exiting. Because gifters may leave a session and return later, our data allow us to observe their full participation histories and capture re-entry behavior. We find that approximately 22% of gifters re-enter a session after exiting. Panel (c) shows the distribution of stay-out length. Conditional on re-entry, gifters remain outside the session for an average of 29 minutes before returning. Panel (d) presents the distribution of session duration for comparison. All four distributions exhibit a roughly exponential shape, motivating our use of exponential arrival and duration assumptions in the structural model.

Session Topics Finally, we briefly introduce the session topics in our data to show that our sample covers a broad range of categories. On TikTok LIVE, creators can include hashtags in their live stream titles. Hashtags—words or phrases preceded by a “#” symbol—serve as searchable keywords that categorize content and enhance its visibility and discoverability. They also assist TikTok’s recommendation algorithm in identifying the nature of the content and matching it to a relevant audience. To identify the topic of each session, we collect the session title and extract any hashtags included. In our sample, approximately half of the sessions include at least one hashtag in the title.

Using these hashtags, we document substantial heterogeneity in content type, audience size, and session duration across sessions. Table .11 ranks topic categories by the number of sessions. The five most popular topics, including #Music, #Daily Life, #Chat&Interview, #Art, and #Dance, account for 80% of all sessions. While some categories, such as Music and Daily Life, account for a large share of total sessions, others, including Beauty & Fashion and Food, attract significantly larger audiences per session. Session durations are relatively stable across most categories, but Gaming and Shopping sessions tend to last considerably longer. These differences highlight the diverse environments in which viewer participation and gifting decisions take place. Appendix Table .11 provides detailed summary statistics by content category.

5 Gifting Motives and Leaderboard Stagnation: Reduced-Form Evidence

This section documents reduced-form evidence on gifting behavior and leaderboard stagnation in livestreaming sessions. It establishes key empirical patterns that motivate the

structural model.

Throughout the paper, we use the term *viewer* to refer to all users attending a livestreaming session. Among them, we define *gifters* as viewers who send a positive amount of gifts during the session. Our empirical and structural analyses focus on gifters, who are the strategic decision-makers in gifting. When the context is clear and passive viewers are not behaviorally relevant, we use the term *viewer* in a generic sense to improve readability.

Using a high-frequency identification strategy, we first establish that gifting behavior is driven in part by competitive incentives: gifters increase their contributions in response to changes in leaderboard rankings in order to improve their standing. We then show that these competitive incentives are not uniform over time or across competitive environments. As leaderboard rankings become more stagnant, characterized by widening score gaps and declining rank mobility, the responsiveness of gifters to ranking changes diminishes substantially. This pattern provides direct evidence that stagnation weakens the effectiveness of leaderboards in sustaining competition. Finally, we distinguish competitive motives from appreciation motives by analyzing gifting behavior among lower-ranked gifters who are unlikely to reach top positions. For these viewers, gifting responds primarily to content popularity, measured by real-time likes, consistent with an appreciation-based motive.

5.1 Identification for Competitive Motive

Identification Challenges To causally test whether gifters value a higher ranking and are willing to gift extra to climb the leaderboard, we need to check whether they will increase their gifting amount if their rankings are exogenously bumped downward. If higher rankings are preferred, after experiencing an exogenous downward shock in their rankings, we expect the gifters bumped downward to increase their gifting, aiming to reclaim their old ranking.

However, conducting such an exercise with observational data presents immense challenges because a clear and sharp exogenous downward shock in ranking is extremely difficult to find. Gifters usually move down the leaderboard only when other gifters out-gift them, which are endogenous actions and could be driven by unobservable contemporary events in the live stream. If a gifter increases her gifting amount after another gifter surpasses her on the leaderboard and moves her ranking downward, it could be either a result of competitive motives to reclaim the old ranking or unobservable events encouraging all viewers to gift simultaneously, e.g., an exciting performance drives a surge of gifts. In extreme cases, even if we could directly collaborate with TikTok to run AB tests to vary the rankings exogenously, the unnatural changes in the rankings without any gifting activities are likely to raise suspicion among gifters, thus contaminating the measurement for competitive motives and biasing our estimates.

High-frequency Identification Fortunately, the design of the leaderboard on TikTok

LIVE provides a distinctive identification opportunity using moment-by-moment data. The leaderboard only ranks gifters who are currently present in a live session, updating in real time to reflect the entry and exit of gifters. Whenever a gifter exits and later re-enters the session, their previous cumulative gifting amount is immediately accounted for and reflected on the leaderboard.

This feature creates a quasi-experimental setting at high frequency. When a top gifter with a large cumulative gifting amount leaves the session for an extended period (more than 10 minutes) and later re-enters, their return bumps down the rankings of all gifters with strictly lower cumulative gifting. For these re-entry events, we refer to the individual re-entering as the *focal gifter* and to the remaining gifters as *non-focal gifters*. We then examine the immediate change in gifting behavior among the non-focal gifters whose rankings are involuntarily demoted by the focal gifter’s return.

High-frequency identification addresses the identification challenges in two ways. First, the re-entry of a focal gifter changes the ranking without involving any new gifting activity, isolating the effect of leaderboard position. Bumped-down gifters face a sudden negative shock to their ranking: their rankings are demoted on the leaderboard, and their rank-associated badges may be downgraded or removed. We could measure the immediate response in gifting behavior towards such adverse shocks to uncover gifters’ preference for maintaining higher leaderboard positions. Second, the key assumption for our identification strategy, that the timing of re-entry is exogenous, is supported by TikTok LIVE’s platform setup. According to TikTok, viewers have no information about the ongoing activities in a live session unless they join it. As such, it is nearly impossible for focal gifters to coordinate their re-entry time with specific activities in the session.

To further address the concerns about re-entry endogeneity, we restrict further to focal gifters who left the session for an extended time, i.e., more than 10 minutes, to minimize their inference about session progression from previous consumption experience. Taken together, these considerations give us confidence that the timing of re-entry is plausibly exogenous. We will systematically validate this assumption using an event study design in the main analysis and a placebo test in the appendix.

5.2 Empirical Design for Competitive Motive

We implement a regression discontinuity design, using time as the running variable around the re-entry event. Such a design is usually adopted in the financial market analysis (Gürkaynak, Sack, and Swanson, 2005; Nakamura and Steinsson, 2018), when continuous financial data is observable, and more recently, a paper in marketing (Liaukonyte, Teixeira, and Wilbur, 2015), when television advertising and online shopping activities are jointly observed in real-time. Our design naturally fits into the above category. The significant difference

is that our regression discontinuity analysis is in conjunction with a narrow window around the event created by a specific viewer, the re-entry time of a top gifter, instead of an event of specific economic or marketing activities. Moreover, we focus on top gifters re-entering instead of exiting for two reasons. First, our data provide precise timestamps for re-entry events, which allow us to clearly identify treatment timing and detect discontinuities in subsequent gifting behavior. Second, exit decisions are made while gifters are actively present in the session and therefore inherently contingent on ongoing within-session activity. In contrast, re-entry decisions are typically made when gifters are outside the session, where observing real-time session dynamics is difficult or impossible, making re-entry substantially less contingent on contemporaneous session activity.

Our regression discontinuity analysis proceeds as follows. For each re-entry event, we measure an individual gifter’s minute-level gifting amount within a six-minute pre-window prior to the re-entry moment. The pre-period serves as a baseline against which the effect of bumped ranking is measured. The same variables are measured again in the sixth minute post-window immediately following the re-entry moment. The systematic difference in non-focal gifters’ gifting behavior between the pre- and post-windows is then attributed to the exogenous variation of the rankings. By focusing on the tight time window around re-entry moments, it is almost surely only changes in rankings that happened exogenously due to the re-entry. Therefore, we could confidently attribute the systematic change in gifting behavior to the effect of the rankings.⁹

We identify all instances in our moment-by-moment data where focal gifters re-entered the session and non-focal gifters who are involuntarily bumped down in their rankings due to those instances. Based on the non-focal gifters’ original rankings, we divide non-focal gifters into those who are bumped from top 1 to top 6.¹⁰ We isolate the 12-minute event windows around the re-entry moment t_0 and run the following regression specifications:

$$\begin{aligned} \text{gift}_{ist} = & \sum_{k=1}^6 \alpha_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k, t \geq t_0\} + \sum_{k=1}^5 \beta_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k\} \\ & + \lambda * \text{others' like}_{ist} + \psi * \text{audience size}_{st} + \phi * t + \theta_s + \delta_{ic(s)} + \epsilon_{ist} \end{aligned} \quad (1)$$

where i indicates the bumped individual, s indicates the session, and t indicates time in minute. Dependent variable gift_{ist} is the total gifts sent by non-focal gifter i in session s at time t . rank_{is,t_0-1} indicates the non-focal gifters’ original rank from which they get bumped down. Independent variable $I\{\text{rank}_{is,t_0-1} = k, t \geq t_0\}$ is a dummy variable equal to one if the

⁹We keep the window relatively short to avoid other potential adjustments. For instance, over 50% of non-focal gifters reclaimed their old rank after the sixth minute, meaning that a wider time window may introduce gifting dynamics after claiming old rank and contaminate the effect of bumped rank.

¹⁰For the clarity of results, we focus on direct bumps when non-focal gifters are directly bumped down by focal gifters who take their position. In other words, we focus on the focal gifter who lands on the k^{th} rank and bumps the k^{th} non-focal gifter from the k^{th} position to $k+1^{th}$ position.

non-focal gifter i was bumped from rank k and the focal gifter had entered the session s at rank k , and others' like_{ist} is the total number of likes sent by viewers other than i in session s at time t to proxy the popularity of the session and control for appreciation motive. We also control the size of viewers who are watching the content of in session s at time t , through audience size $_{st}$, the number of minutes since the start of the session t , session fixed effect θ_s , and viewer-creator fixed effect $\delta_{ic(s)}$, where $c(s)$ is the creator for session s . Specifically, audience size $_{st}$ controls for the effect of viewer size on gifting motive (Lu et al., 2021), and also controls for the fact that audience size is positively related to a larger number of likes sent, and t captures that the sessions often become more or less popular over time due to the natural life progression of sessions. Session fixed effects θ_s capture unobserved characteristics of sessions that affect the viewers' incentive to gift, such as whether a popular creator holds the session or whether the session is during holidays. Finally, the viewer-creator fixed effects $\delta_{ic(s)}$ capture unobserved characteristics of individuals that affect their incentive to gift to a specific creator, such as income level or how much they like specific creators. The error terms can be correlated within a session by allowing clustered standard errors at the session level. This is because the re-entry event is at the session level, and the gifting behavior in the same session is likely to be correlated across viewers and time.

5.3 Evidence of the Competitive Motive

This section presents the evidence for the competitive motive estimated from the regression equation (1). We first show that the leaderboard generates a competitive motive in gifting activities - a preference for ranking by viewers.

Table 2: Change in Gifting Behavior for Bumped-down Viewers

	Gift by bumped-down viewers					
	1 to 2	2 to 3	3 to 4	4 to 5	5 to 6	6 to 7
$\alpha_{k,k+1}$	3.4688** (1.4157)	1.4125 (2.0483)	3.0476** (1.3012)	1.5827 (1.0224)	1.4376 (1.7741)	0.3505 (0.5943)
λ	0.0085*** (0.0018)					
Observations	261,948					
Controls	YES					
Session FE	YES					
Viewer $\times$ Creator FE	YES					

Note: This table presents the evidence for the competitive motive estimated from the regression equation (1). Controls and fixed effects include the size of the viewers of the session, audience size $_{st}$, the number of minutes since the start of the session t , session fixed effect θ_s , and individual times creator fixed effect $\delta_{ic(s)}$. Standard errors are in parentheses. Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level.

The Presence of the Competitive Motive We find a rank-ordered preference by view-

ers, which is closely related to the reward scheme to reward the top-three gifters with badges to play along with their name on the leaderboard and comment section. More specifically, the particular preference is as follows:

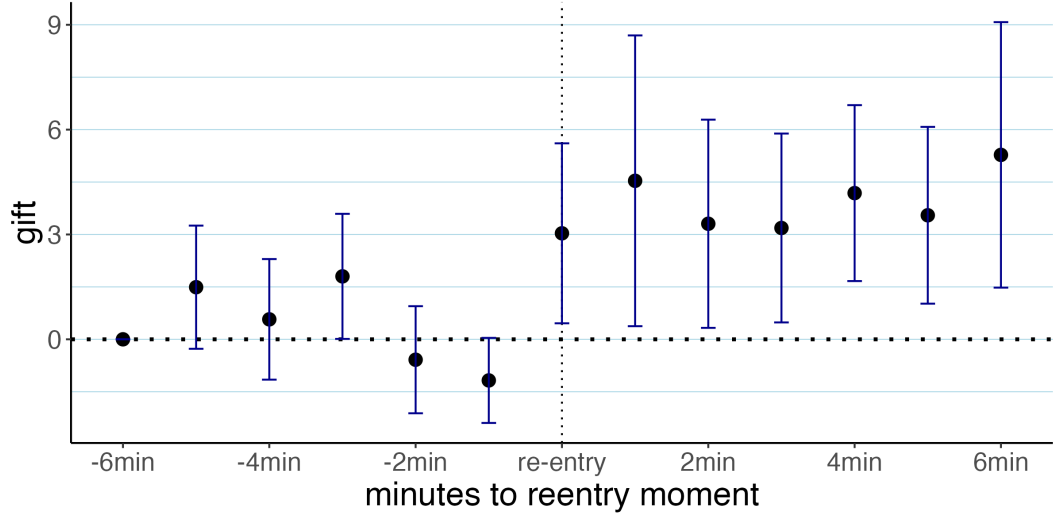
$$\text{rank 1} \succ \text{rank 2} \sim \text{rank 3} \succ \text{rank 4} \sim \text{rank 5 and beyond} \quad (2)$$

which is shown in the regression results in Table 2. More specifically, the table shows the change in gifting behavior of viewers when their ranking is bumped down by one due to the re-entry of focal gifters directly above them, i.e., a rank 3 gifter is bumped to rank 4 due to the re-entry of a focal gifter who claims a new rank 3 position. We find significant increases in gifting amount when a gifter gets bumped down from the rank 1 to the rank 2 (column 1), and rank 3 to the rank 4 (column 3). However, we find a positive but insignificant change in gifting amount when bumped down from other rankings. This indicates an ordered preference among gifters for ranking on the leaderboard: viewers strictly value rank 1 over rank 2 and rank 3 over rank 4, but are indifferent between rank 2 and rank 3. When viewers get bumped down from top 4 to top 5, or from any ranking beyond rank 5, the change in gifting amount is insignificant, suggesting viewers are indifferent between those rankings. I do not present any results after the top 6 because those are all insignificant, following the logic.

The results in Table 2 and Equation (2) exhibit the direct impacts of the leaderboard design of TikTok LIVE on the competitive motives of viewers. The first key cutoff is at the No.1 position, with the highest visual prominence. Viewers place a high value on the No. 1 position due to its visual prominence at the top of the leaderboard and the "No.1" badge when commenting. As shown in our institutional background. The second key cutoff is at the No. 3 and No. 4 positions, where the badge is on the margin of being lost or gained. Losing the badge at the No. 3 position generates a strong desire among viewers bumped from the third position to reclaim their original spot and badge. Finally, we find no evidence that leaderboard visibility alone generates additional preference for higher rank. Although the leaderboard introduces a visibility threshold between rank 5 and rank 6, where only the top five are immediately visible, being bumped from rank 5 does not lead to a significant increase in gifting, indicating that visibility per se does not drive competitive incentives.

We further validate the results in Table 2 with an event study in Figure 3 to address the worry that focal gifters strategically choose their re-entry time to coincide with significant in-live events, such as highlighted performance or thrilling challenges. The regression

Figure 3: Gift of Bumped-down Gifters Around the Re-entry Moments



Note: This figure plots the regression coefficients $\alpha_{\delta t}$ from a linear model that regress the gift of bumped viewers (from rank 1, 3) on a set of dummies to capture one-minute intervals within 12-minutes time span around the re-entry moment of focal gifters, while setting minus-six minute as the baseline and controlling for number of minutes since the start of session, others' likes, audience size, session fixed effect, and viewer $\times$ creator fixed effects. The error bar is at a 10% confidence interval.

specification is as follows:¹¹

$$\begin{aligned} \text{gift}_{ist} = & \sum_{\Delta t=-5}^6 \alpha_{\Delta t} * I\{t = t_0 + \Delta t\} + \lambda * \text{others' like}_{ist} \\ & + \psi * \text{audience size}_{st} + \phi * t + \theta_s + \delta_{ic(s)} + \epsilon_{ist} \end{aligned}$$

If the assumption is violated, we expect gift by viewers to smoothly increase even before the focal gifter re-enters the live session, reflecting that the session is already trending up before the re-entry moment. Figure 3 shows that the gift of bumped viewers remain unchanged before the stream and increase immediately after the re-entry moment. Therefore, we do not find any indication that endogeneity from re-entry time poses a substantial threat to our empirical strategy. For the full regression specification and results of the event study, please refer to Section B.1 in the Appendix. Finally, we conduct an additional robustness check on the assumption of exogenous re-entry timing through a placebo test. In Appendix B.2, we replicate the baseline specification by assigning a fictitious re-entry time 10 minutes prior to the actual re-entry event and examining gifting behavior around this placebo cutoff. We find no discontinuity or systematic change in gifting at the placebo timing, indicating that the observed effects arise specifically at the true re-entry moment rather than from pre-existing trends.

Moving beyond the average preference for rankings above, we want to understand how

¹¹For full regression result, please refer to Appendix B.1.

viewers’ competitive motives vary across different live-stream conditions. This leads us to explore heterogeneity in competitive motives at varying levels of competition difficulties faced by non-focal gifters in the next section.

Table 3: Heterogeneous Effect of Gifting Distance to Climb Up

	Gift by bumped-down viewers		
	1 to 2	2 to 3	3 to 4
$\alpha_{k,k+1,low}$	1.3498* (0.7564)	2.8546 (2.6586)	3.4395** (1.4640)
$\alpha_{k,k+1,median}$	8.3763*** (2.7300)	-0.5391 (0.6826)	1.4329* (0.7832)
$\alpha_{k,k+1,high}$	0.0949 (4.6817)	1.8538 (3.4039)	1.1229 (2.4649)
λ		0.0094*** (0.0026)	
Observations		198,996	
Controls		YES	
Session FE		YES	
Viewer $\times$ Creator FE		YES	

Note: This table presents the evidence for stagnation in the competitive motive. This analysis exploits the re-entry event to examine how the difficulty of overtaking higher-ranked competitors moderates gifters’ competitive incentive. We proxy the difficulty of regaining a prior rank by the score gap, i.e., the difference in cumulative gifting amount between focal gifter and non-focal gifter. We find the bumped non-focal gifter only significantly increase their gifting amount when the score gap to focal gifter is either in the low or median range. Once the score gap is large, they do not significantly increase their gifting amount, indicating discouraged competitive incentive. This serves as direct evidence of the stagnation. Standard errors are in parentheses. Significance: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors are clustered at the session level.

5.4 Evidence for Stagnation

This section provides evidence of stagnation on the leaderboard. We examine how the difficulty of overtaking higher-ranked competitors moderates viewers’ competitive incentives. Specifically, exploiting the same re-entry events, we conduct a heterogeneity analysis of gifters’ competitive responses. We proxy the difficulty of regaining a prior rank by the score gap, i.e., the difference in cumulative gifting amounts between the focal gifter and the non-focal gifter immediately before the re-entry event, and study how the non-focal gifter’s

response varies with this gap. The specification for the heterogynous analysis is as follows:

$$\begin{aligned}
\text{gift}_{ist} = & \sum_{k=1}^3 \sum_{d \in \{\text{low}, \text{median}, \text{high}\}} \alpha_{k,k+1,d} * I\{\text{rank}_{is,t_0-1} = k, \text{dist}_{is,t_0-1} = d, t \geq t_0\} \\
& + \sum_{k=1}^2 \sum_{d \in \{\text{low}, \text{median}, \text{high}\}} \beta_{k,k+1,d} * I\{\text{rank}_{is,t_0-1} = k, \text{dist}_{is,t_0-1} = d\} \\
& + \sum_{d \in \{\text{low}, \text{median}\}} \beta_{3,4,d} * I\{\text{rank}_{is,t_0-1} = k, \text{dist}_{is,t_0-1} = d\} \\
& + \lambda * \text{others' like}_{ist} + \psi * \text{audience size}_{st} + \phi * t + \theta_s + \delta_{ic(s)} + \epsilon_{ist}
\end{aligned}$$

Here, $k \in \{1, 2, 3\}$ denotes the non-focal gifter's rank immediately before the re-entry event at $t_0 - 1$. Within each rank group, we further classify non-focal gifters into low, median, and high score-gap categories based on the distance between their cumulative gifting amount and that of the focal gifter. The score-gap thresholds correspond to the 33rd and 66th percentiles. In the regression, dist_{is,t_0-1} captures the non-focal gifter's score distance relative to the focal gifter just prior to re-entry.

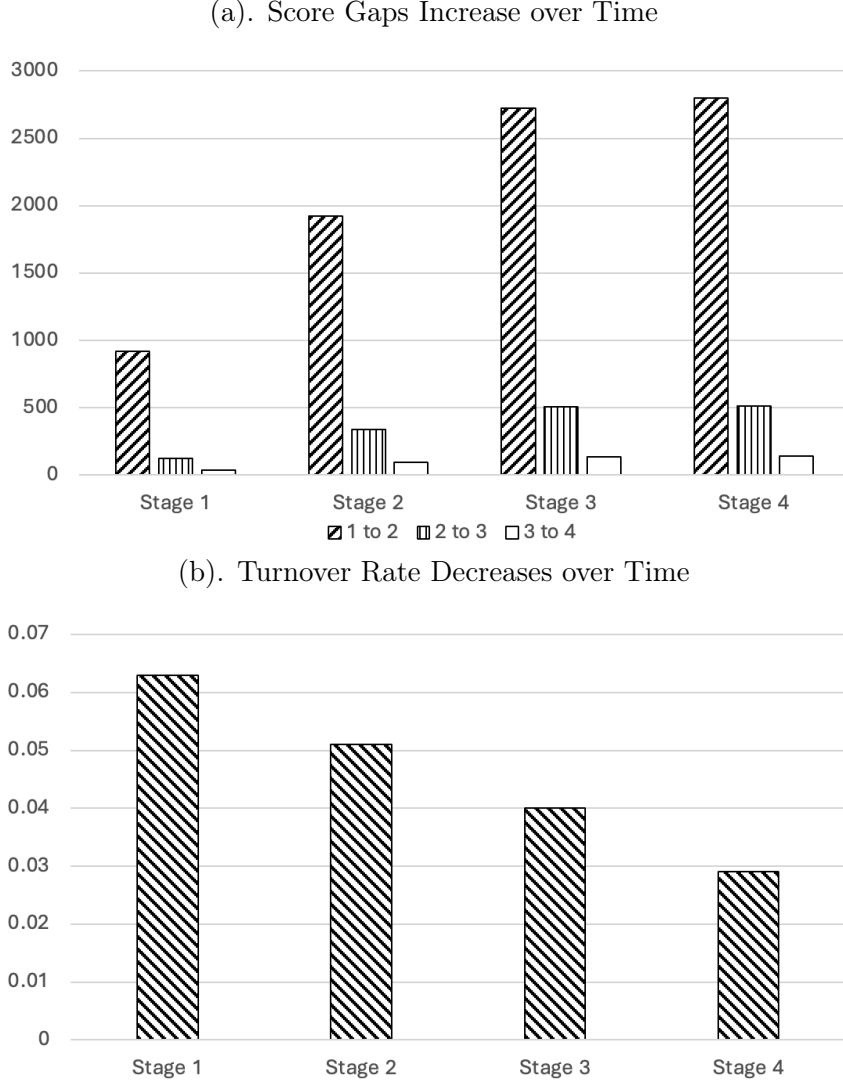
Table 3 shows the evidence of leaderboard stagnation. We find that viewers bumped from ranks one and three only significantly increase their gifting amount after re-entry when the score distance is low or median. When the distance becomes large, they do not significantly increase their gifting amount, indicating that they become discouraged from competing when reclaiming their prior rank is sufficiently costly. Together, these results provide direct evidence of a stagnation problem: as score gaps widen, competitive incentives weaken, and bumped viewers become less responsive to ranking shocks.

Moreover, we find that stagnation intensifies over the course of a livestreaming session. We measure stagnation using two complementary metrics. First, for each $k \in \{1, 2, 3\}$, we compute the score gap between the viewer ranked k and the viewer ranked $k+1$. Larger gaps indicate greater difficulty for lower-ranked viewers to overtake those above them. Second, we compute the turnover rate of the top-three positions—the frequency with which the identity of the top-three players changes due to gifting competition, measured per minute. Lower turnover reflects greater stagnation, as it indicates that top positions become increasingly locked in. As shown in Figure 4, stagnation rises sharply over time: score gaps between adjacent ranks widen substantially (top panel), while the turnover rate of the top three steadily declines (bottom panel).

5.5 Evidence of the Appreciation Motive

Finally, we present evidence of the appreciation motive, such as viewers sending gifts to express their appreciation for high-quality content. This motive is a spontaneous reaction

Figure 4: Stagnation Increases over Time During the Session



Note: This figure illustrates the evolution of stagnation over the course of livestreaming sessions using two measures: score gaps between adjacent ranks (top panel) and turnover rates among the top-three positions (bottom panel). Each session is divided into four equal time stages, and we compute the average score gaps and turnover rates within each stage. We observe a consistent pattern of increasing stagnation over time: score gaps widen as sessions progress, while turnover among top positions steadily declines.

to content quality, requiring no strategic interaction with other viewers in the sessions. For a typical viewer i , we use the total number of likes sent by all other viewers in the same session to proxy the content's popularity at each minute of the session and investigate how the quality of the content affects the gifting motive of viewer i .

Our evidence comes in two ways. First, in the previous analysis in both Tables 2 and 3, we find that, besides the competitive motive, top-ranked gifters also gift out of appreciation. They show positive responses to gifting in terms of the total number of likes from others. Second, we further investigate the appreciation motive by restricting our sample to lower-ranked gifters (who never ranked top 4 or above throughout the session they attend) to rule

Table 4: Appreciation Motive for Gifting

	gift
others' like	0.0012*** (0.0001)
audience size	0.0011*** (0.0004)
t	0.0003* (0.0001)
Observations	13,591,314
Session FE	YES
Viewer $\times$ Creator FE	YES

Note: Standard errors are in parentheses. Significance: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors are clustered at the session level.

out the potential influence of the competitive motive. Since these viewers are not responding to re-entry events, we run the following regression to only examine their appreciation motive:

$$\text{gift}_{ist} = \lambda * \text{others' like}_{ist} + \psi * \text{audience size}_{st} + \phi * t + \theta_s + \delta_{ic(s)} + \epsilon_{ist}$$

We discard the re-entry timing and rank dummy in equation (1), while keeping all other controls and fixed effects the same. Table 4 shows that the popularity of content, proxied by the total likes by others, positively increases the total gifts of these viewers who do not show any competitive motives, supporting the presence of the appreciation motive. Both results show that the motive for appreciation is universal across all viewers.

This section provides reduced-form evidence on two distinct motives underlying gifting behavior in livestreaming sessions: a competitive motive induced by leaderboard rankings and an appreciation motive driven by content quality. The empirical analyses establish that viewers actively respond to leaderboard-based competition and reveal a separate, non-competitive channel through which gifting reflects appreciation for high-quality content. These findings motivate the structural model, which explicitly incorporates both motives to quantify their relative importance and economic magnitude. In addition, this section documents systematic patterns of leaderboard stagnation, showing that competitive incentives weaken as rankings become increasingly difficult to overturn. These empirical patterns provide the foundation for the counterfactual analyses in subsequent sections, in which we evaluate alternative leaderboard designs to manage stagnation and sustain competition.

6 A Continuous-Time Structural Model of Gifting Competition

In this section, we build on the reduced-form evidence and develop a dynamic structural model of gifting competition. The model serves three purposes. First, it allows us to estimate key preference primitives, including viewers’ valuation of top leaderboard rankings and their appreciation for high-quality content. Second, it enables us to quantify the economic impact of leaderboards on session-level revenue across different session characteristics. Third, the model provides a framework for evaluating alternative leaderboard scoring designs, allowing us to assess how redesigning the scoring system can mitigate stagnation, sustain competition, and improve platform revenue.

6.1 Overview of the Model

In this section, we develop a structural model of gifting behavior, building on the insights from the reduced-form analysis presented. We model gifters’ behavior within a livestreaming session as a dynamic game, in which viewers compete over time for a limited number of positions on the session-specific leaderboard. Decisions are dynamic because gifters are forward-looking and take into account how current gifting affects their future ranking positions on the leaderboard. For example, by contributing earlier in the session, a gifter may accumulate a score advantage that increases the probability of retaining a top rank over subsequent periods. We model play using a continuous-time, rather than a discrete-time game, for several reasons. The first is realism: sessions evolve continuously, and viewers’ decisions can occur at any instant during play. Assuming discrete play involves arbitrarily breaking observed sessions into a finite number of discrete intervals ([Heckman and Singer, 1986](#)). The second reason is that the formulations of continuous-time games used in empirical work, e.g., [Arcidiacono et al. \(2016\)](#), have appealing theoretical and computational properties. A pivotal assumption in many empirical applications of continuous-time games, which we employ, is that the timing of agents making decisions is determined by a continuous process. Importantly, this assumption implies that the probability of two or more agents making decisions simultaneously is zero. In discrete-time games, simultaneous decisions can lead to multiple equilibria. The fact that this source of multiple equilibria is ruled out makes computing counterfactuals much simpler ([Blevins and Kim, 2024](#)). An additional computational advantage of continuous-time games is that, since only a single agent can make a decision at any instant, solving for the equilibrium is computationally easier. In a discrete-time game, one must iterate over each agent’s best responses to find an equilibrium, increasing the solution time exponentially as the number of players increases.

Following the literature outlined previously on continuous-time models, we assume that

the instant at which a player may make a decision, as well as the evolution of state variables (e.g., content quality and session end), is governed by a continuous process that we estimate. Thus, we model gifters’ entry and re-entry with exogenous continuous processes. By exogenous, we mean that the entry and re-entry probabilities are modeled as functions of observable states, and do not depend on variables that are unobserved to the researcher, but observed to gifters. This assumption is informed by the way TikTok LIVE is set up. Gifters cannot access any information inside a session—such as the content quality, leaderboard rankings, or other gifters’ gifting behavior—until they enter or re-enter the session. This lack of outside visibility limits their ability to make informed entry decisions based on the current state of a session. As a result, we model entry and re-entry as being driven by external factors (e.g., the day of the week, or session-level variables that may enter TikTok’s matching algorithm), rather than strategic considerations about in-session dynamics. This assumption allows us to treat the entry flow as exogenous to the gifting decision process modeled within the session. We assume that the entry and re-entry times are determined by exponential distributions. This assumption is consistent with the empirical distributions of entry and re-entry timing presented in Figure 2, panels (a) and (b), which can be seen to have a shape similar to an exponential distribution.

In the gifting decisions, viewers decide how much they want to send to the content creators. We allow their gifting decision to be driven by two motives, the competition and appreciation motives, as suggested by our reduced-form evidence in Section 5. The competition motive stems from the fact that we observe viewers are willing to gift more to recapture the top spot after they are exogenously moved out of those slots. Additionally, our model captures the stagnation by viewers’ tradeoff in competing for higher positions. Our previous reduced-form finding suggests a negative relationship between a viewer’s competitive motive and the score gap to its adjacent players on the leaderboard. This finding suggests that gifters trade off the cost of making a gift with the benefit of improving their rankings. If it would cost too much to claim the top spot on the leaderboard, they do not incur the expense. The appreciation motive stems from the finding that viewers are observed to gift more when the number of audience likes is higher, controlling for the overall size of the audience. In our structural model, we incorporate these motives into viewers’ utility functions. The gifters in the game receive higher utility from being in the top spot on the leaderboard, i.e., obtaining top-3 badges. Since only one gifter can occupy a given slot, gifters will have an incentive to compete for more valuable positions. We model the appreciation motive by assuming the marginal utility of giving a gift is an increasing function of overall content quality. In the model, we also allow consumers to continuously generate consumption value from watching live content, even if they do not gift.

In addition to the gifting decision, a gifter can also make endogenous exit decisions. They will appropriately trade off the utility of continuing to accrue payoffs from gifting in

the session and enjoying its content, or exiting the session with the possibility of re-entering later.

At any point in time, we assume that up to N gifters can be active in the game. An active gifter is a user who is attending the livestream at a given instant and makes at least one gifting decision during the session. In the data, only a small fraction of viewers, less than one percent ever send gifts, implying that strategic interaction is concentrated among a limited set of participants. Accordingly, we model the behavior of active gifters as strategic players and abstract from the behavior of the remaining viewers.

Viewers who attend the session but never send gifts are treated as passive and non-strategic agents. While these passive viewers do not participate in gifting competitions, their interactions with the livestream, such as pressing the “like” button, provide information about content quality. We therefore model the evolution of likes as an exogenous state variable capturing time-varying content quality. The total number of likes includes contributions from both active gifters and passive viewers.

For computational tractability, we restrict the maximum number of active gifters to $N=7$ in estimation and counterfactual analyses. This restriction applies at each point in time and is unlikely to affect our results, as sessions in which more than seven gifters are simultaneously active account for less than one percent of total gifting revenue in the data.

6.2 Entry, Re-entry, Content Quality and Session Termination

We now describe how we model the evolution of gifter participation over time, including gifter entry and re-entry, as well as the exogenous processes governing content quality and session termination.

We model the entry time into a live session as a random variable, τ_e , drawn from an exponential distribution with hazard rate $\lambda_e(W_t^e) = (W_t^e)' \boldsymbol{\theta}^e > 0$, where W_t^e is a vector of session-level characteristics and $\boldsymbol{\theta}^e$ is a vector of parameters. Figure 2(a) presents the empirical distribution of entry times, which closely follows an exponential distribution, consistent with our distributional assumption.

The reduced-form treatment of entry reflects the information structure of TikTok LIVE. Viewers make entry decisions while outside the live session and therefore cannot observe within-session competitive states, such as leaderboard positions or score gaps, prior to entry. As a result, entry timing cannot be strategically optimized based on contemporaneous competition and is instead modeled as a stochastic process that captures systematic variation in exposure and availability. Accordingly, we include two covariates in W_t^e : indicators for the day of the week and a proxy for real-time session content quality, measured by total likes. Day-of-week indicators capture routine-driven participation and responsiveness to live-stream notification, reflecting systematic variation in viewers’ availability over the week.

Real-time likes proxy for session popularity and exposure, capturing the likelihood that a session is surfaced to viewers through TikTok’s recommendation system, rather than viewers’ strategic anticipation of competition within the session. Because our analysis conditions on gifters, who are overwhelmingly followers of the creator, we abstract from the follower margin and focus on participation dynamics conditional on this population.

Conditional on exiting the session, the time a gifter remains outside before re-entry, τ_{re} , is also modeled as a random variable drawn from an exponential distribution with hazard rate $\lambda_{re}(W_t^{re}) = (W_t^{re})'\boldsymbol{\theta}^{re} > 0$.¹² The re-entry hazard depends on a vector of session-level characteristics, W_t^{re} , including indicators for the day of the week and a proxy for real-time session content quality. Our reduced-form regression analysis in Section 5 supports this specification. In particular, we find no evidence of pre-trends in gifting behavior prior to re-entry events, suggesting that re-entry timing is orthogonal to within-session strategic incentives such as leaderboard scores. Figure 2(c) shows that the empirical distribution of re-entry times is well approximated by an exponential distribution, consistent with our modeling assumption.

At any moment of the session, we call the gifters who are present in the session *incumbents*, and the gifters who have exited the session *idle gifters*, which we capture with a state variable $g_t \in \{0, 1\}$. At any point in time, if a gifter is in the active state $g = 1$, she is present in the session; otherwise, she has exited the session and has not re-entered yet. At any moment of the session t , a gifter’s expectation about the future number of incumbents depends on the rate of entry (λ_e), the rate of re-entry (λ_{re}), the number of incumbent gifters, N_t^{inc} , the number of idle gifters, N_t^{idle} , and the policy functions describing an incumbent’s exit decision, which we will present in more detail in the next section. We assume entry and re-entry are only possible when the number of active incumbents does not exceed the maximum number of active gifters, i.e., $N_t^{inc} < N$. We assume that gifters understand this restriction and internalize it in their expectations.

We proxy time-varying content quality using the number of likes received in a given minute of the session. Likes are a natural measure of contemporaneous audience engagement and are highly correlated with other indicators of content quality, such as comments and viewership growth.

For computational tractability, we discretize content quality into two levels, $l_t \in \{l^{low}, l^{high}\}$, representing relatively low- and high-quality moments within a session.¹³ The evolution of

¹²Most gifters do not re-enter after exiting. The fractions of gifters who re-enter 0, 1, 2, and 3 times are 76%, 15%, 5%, and 2%, respectively.

¹³The numerical values of l^{low} and l^{high} are constructed by normalizing minute-level likes relative to the session median to ensure comparability across sessions. Details are provided in Appendix C.1.

content quality follows a continuous-time Markov jump process with intensity matrix Q_q .

$$Q_q = \begin{pmatrix} -q_{12} & q_{12} \\ q_{21} & -q_{21} \end{pmatrix}. \quad (3)$$

where $q_{12}(q_{21})$ is the transition rate from l^{low} to l^{high} (l^{high} to l^{low}). Importantly, we assume that content production, including both session termination and time-varying content quality, is exogenous to viewers' actions. In particular, gifting does not directly affect the evolution of content quality during the session. This specification reflects the idea that, for the mature creators in our sample, content production decisions are largely predetermined at the session level rather than adjusted in real time in response to gifting. We provide supporting empirical evidence for this assumption in Appendix D.2.

The session termination is modeled as a two-stage process, where the transition is governed by a latent state $h_t \in \{0, 1\}$. We assume that the session starts with the latent state off, $h_t = 0$, and that at rate q_{latent} the latent state switches on, i.e., $h_t = 1$. When the latent state is off, the session will not end. Once the latent state is on $h_t = 1$, the session ends at rate q_{live} . The latent state h_t is observable to the viewers in the session but not observable to us as researchers.

The two-stage process captures the TikTok setup. On TikTok, there is no scheduling tool on the platform to help the content creator schedule the session ahead of time, so the session duration remains stochastic to viewers. In expectation, the session enters the end stage with a probability that the creator announces it is wrapping up, and the session ends soon with high probability. We note that our stochastic process for the session termination assumes that it is exogenous to gifters' actions, such as gifting.

6.3 Gifter Utility and Decisions

In this section, we develop the structure of the gifters' dynamic problem. We model two endogenous decisions that gifters can make: how much to gift and whether to exit. Following the previous literature on empirical continuous-time games such as Arcidiacono et al. (2016), we assume that the time at which a gifter can make a decision is stochastic and follows a Poisson process with rate λ_a . Gifters can choose between exit, stay and gift a positive amount, or stay and gift nothing when they are given an opportunity to make a decision. This assumption stands in contrast to discrete-time games, where it is usually assumed that all players may make decisions at any discrete time period. As described in Section 6.1, the assumption that decision points arrive stochastically rules out the possibility that agents make simultaneous decisions, simplifying the computation of equilibrium relative to discrete time games.

States Variables

Gifters make their exit and gifting decisions based on the current state of the session and their beliefs about the evolution of the session’s state variables. We now describe each state variable in the model. The first two variables capture player scores. In TikTok LIVE streams, a gifter’s score reflects her cumulative gifting amount. Recall from Section 3 that players purchase TikTok coins for a price of about 1.4 cents per coin. The leaderboard continuously updates how many coins has been contributed by each gifters during the live stream, and resets the record after the session is over. In the model, we discretize the scores into $K + 1$ different levels: $s_{it} \in \mathbb{S} = \{score_0, score_1, \dots, score_K\}$.¹⁴ A score of $s_{it} = score_0 = 0$ means the gifter has not gifted yet within the current session.

We assume that gifters track both their own score, s , as well as the distribution of all incumbents’ scores, $D_t = (d_0, d_1, \dots, d_K)$, where d_k is the number of gifters whose score is at level $score_k$. Importantly, gifters are assumed to condition their decisions only on the distribution of scores rather than on the identities of individual opponents. This assumption reflects the institutional features of livestreaming leaderboards, which prominently display ranks and scores but provide limited information about the identities or past behaviors of other viewers. As a result, competitive incentives are driven by relative standing on the leaderboard rather than by strategic considerations toward specific opponents.

Gifters also track their own rank, $r_t \in \{1, 2, 3, 4\}$. Note that this state is not redundant with the score, because the score is discretized. Consistent with TikTok’s setup, if two gifters have the same score, the model lets the gifter who arrived first at a given level have a higher rank. Because of the underlying discretizations, the rank must be tracked as a separate state. We assume gifters track whether they are rank 1, 2, 3 or 4. A gifter is rank 4 if they are not among the top 3.

Besides the three state variables described in the previous paragraph, gifters also track the following three state variables: First, whether the gifter is idle or not, which is captured by g_t . Recall that $g_t = 0$ if a gifter is idle (has exited and not re-entered yet) and $g_t = 1$ if they are active in the session. Second, they also track the content quality, l , which follows an exogenous Markov process, as described in the previous section. Third, they track the total number of idle gifters, N_t^{idle} . For computational tractability, we assume that players do not track the scores of idle gifters. We will describe how we model expectations about the scores of idle gifters who re-enter later in this section. Fourth, whether the latent state governing the session ending is on or off, which is captured by $h_t \in \{0, 1\}$.

Utility

Next, we describe the structure of gifters’ utility functions. A gifter’s utility from taking an action (gifting, exiting, or doing nothing) has two components: an instantaneous utility, which is immediately incurred on taking the gifting action, and a flow utility that is received continuously until another gifting action happens. The flow and instantaneous utility a gifter

¹⁴I choose grid points of 0 100, 300, 600, 1000 and 1500.

gets from gifting a particular amount and/or staying in the session are functions of three state variables: their own score, s_t , their rank, r_t , and the content quality, l_t . If a gifter decides to make a gifting decision, we denote the gifter's score after gifting to be s'_t . If a gifter sends a positive amount, her score will increase by a positive amount $s'_t - s_t$. Motivated by the fact that scores are discrete, we model the action of gifting as a discrete action: the gifter can change her score to any level which is weakly greater than the current score, i.e., $s'_t \geq s_t$. When a gifter makes a gifting decision, she receives an instantaneous utility as follows:

$$u^{inst}(s'_t, s_t, l_t) = \underbrace{(\gamma_0 + \gamma_1 \cdot l) * \log(1 + 0.01 * (s'_t - s_t))}_{\text{appreciation utility}} + \underbrace{\beta_p \times 0.01 * (s'_t - s_t)}_{\text{gifting disutility}}, \quad (4)$$

where the first term represents her utility from showing appreciation, and the second term represents her disutility from paying for the gifts, i.e., β_p is the price coefficient. Notice that we normalize the gifting amount to hundreds when it enters the utility. We model the appreciation motive by allowing the marginal utility of giving a gift to be an increasing function of the overall content quality, i.e., $\gamma_1 > 0$. This assumption is motivated by the fact that we find empirically that gifts increase when total likes in the audience are higher (Table 4). It is important to note that the appreciation utility contains two terms: the first term, γ_0 , captures the baseline marginal utility from showing appreciation, and γ_1 captures how the marginal utility from showing appreciation varies with content quality. The concave specification reflects diminishing marginal returns to expressing appreciation, while the linear disutility term captures the constant marginal cost of spending on virtual gifts under the platform's pricing scheme.

We model the flow utility of being in a session to be a function of the gifter's rank, r_t , and the number of likes, l_t . The specification we use for the flow utility is as follows:

$$u^{flow}(r_t, l_t) = \underbrace{u_{r_t}}_{\text{rank-based utility}} + \underbrace{\gamma_2 + \gamma_3 \cdot l_t}_{\text{utility from watching live}} \quad \text{where } u_1 > u_2 = u_3. \quad (5)$$

The first term u_{r_t} is the most important part of the model, capturing the gifters' social value of being at rank r_t . Motivated by the reduced-form evidence, we assume that $u_1 \geq u_2 = u_3 \geq u_4 = 0$. Moreover, rank-based utility is a flow utility because gifters value the process of holding the top-gifter badges, during which they receive social utility of enhanced social recognition or a higher chance of interacting with content creators. The second term captures a gifter's value of watching and participating in a session: γ_2 measures the baseline utility from watching content, and γ_3 captures how the utility from consuming content varies with content quality. If a gifter decides to exit, she receives a flow utility of 0. Whenever an agent makes a choice, her instantaneous utility also receives an additive choice-specific shock,

$\varepsilon_{i,s'-s,t}$ for a choice of gift amount, and $\varepsilon_{i,e,t}$ if she exits. We assume that the choice-specific error follows a Type 1 extreme-value distribution.

Finally, we summarize several key modeling assumptions and discuss supporting empirical evidence presented in Appendix C. In particular, we examine alternative explanations for gifting behavior—including herding, free-riding, end effects, and reciprocity-based motives—and find limited empirical support for these channels. We also validate the use of total likes as a proxy for content quality by demonstrating its strong association with other measures of viewer engagement.

In addition, we assess the specification of the appreciation motive as an instantaneous utility that depends on contemporaneous content quality. This specification captures the idea that gifting serves as an immediate expression of appreciation for the content being consumed rather than a delayed response to accumulated viewing experience. Empirically, gifting responds primarily to current content quality, with little sensitivity to lagged quality measures (Appendix C.9). Moreover, we test for potential interactions between competitive and appreciation motives and find no economically meaningful interaction between the two. Together, these results provide support for the modeling choices adopted in the main analysis. Finally, While the model focuses on within-session competition and abstracts from substitution across sessions, Appendix C.10 provides reduced-form evidence examining whether leaderboard lagging leads viewers to substitute toward other sessions.

7 Equilibrium Concept

We model gifter behavior within a livestreaming session as a continuous-time dynamic game. Although the game unfolds over time, the session ends stochastically and the environment is stationary: payoffs and state transitions do not depend explicitly on calendar time. As a result, we focus on stationary equilibrium behavior.

We characterize equilibrium using a Markov Perfect Equilibrium (MPE). In an MPE, gifters choose gifting and exit actions optimally as functions of the current payoff-relevant state, taking as given the Markov strategies of other gifters. The state variables—such as score, rank, and session status—are publicly observable, and there is no private information or belief updating about hidden types. Consequently, equilibrium strategies depend only on the current state rather than on the full history of play. Accordingly, we suppress explicit time subscripts in value functions and strategies throughout the remainder of the analysis.

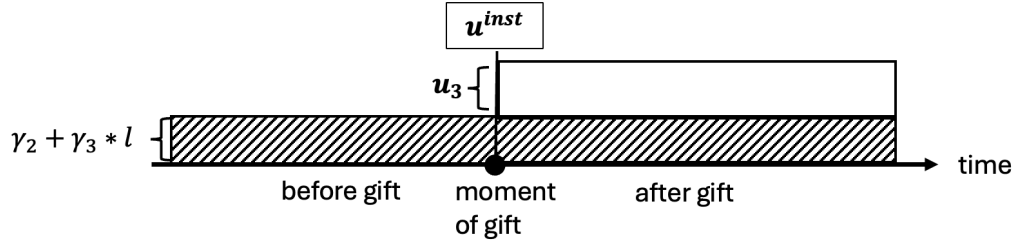
8 Dynamic Optimization and Bellman Equation

The decision of a gifter to give a certain amount, exit, or do nothing are the outcome of a continuous time dynamic game. The state of the dynamic game may change if a gifter has an

opportunity to make a decision and gifts or exits, a new gifter enters, an idle gifter re-enters, or an exogenous variable changes. Importantly, we assume that gifters are forward-looking. This means that when a gifter has an opportunity to make a decision, that decision will depend on both the instantaneous utility, as well as the flow utility and how she expects the state of the game to evolve over time.

Figure 5 illustrates the short-term and long-term tradeoffs faced by a viewer when making a gifting decision. In the figure, the viewer considers whether to gift in order to improve her ranking to the No. 3 position. At the moment of gifting, she faces an immediate tradeoff between the disutility of spending and the utility from expressing appreciation for the content. Beyond this instantaneous tradeoff, gifting also affects the viewer's continuation value by improving her ranking from fourth place to third place, which confers increased visibility and social recognition through leaderboard badges. At the same time, a higher ranking may change the competitive incentives faced by other viewers, shaping subsequent gifting dynamics. The optimal gifting decision therefore balances immediate costs and benefits against the expected long-term gains from achieving and sustaining a higher ranking position.

Figure 5: Short-Run and Long-Run Tradeoffs in Gifting Decisions



Note: The horizontal axis represents time within a livestreaming session. The black dot indicates the moment at which the viewer makes a gifting decision. The shaded area represents the flow utility from watching the content, which depends on content quality. The vertical installment, u^{inst} , at the moment of gifting represents the instantaneous utility from paying the gifting cost and expressing appreciation for the content. The unshaded area after gifting illustrates the higher flow utility associated with obtaining the No.3 badge.

As the examples above illustrate, gifters' expectations will be determined by the model's states and their transition process. We denote the state vector as $k = (g, s, r, D, l, N^{idle}, h)$, where (g, s, r) are the gifter's own states, (D, l, N^{idle}, h) are public states. g, s , and r are state variables because they directly enter the gifter's flow utility function, and thus influence their decisions to gift or exit. Similarly, l also directly impacts utility, and thus gifting/exit decisions. D is a state variable because it impacts a gifter's flow utility through its effect on rank. A gifter may be less likely to gift, for example, if the top slot on the leaderboard is occupied by someone who has a much higher score. N^{idle} is a state because it impacts the likelihood an idle gifter re-enters, and hence the future competitiveness of the game and the value of gifting or exiting. As we will describe in more detail below, the likelihood of re-entry increases with the number of idle gifters. Finally, h is a hidden state that is unobservable

by the researcher, which governs whether the session enters the end stage.

State Transition The state transition function $k'(k; ev)$ describes how the payoff-relevant state evolves following an event ev . Events affect only a subset of state variables, and the transition structure is stationary and Markovian. From the perspective of a focal incumbent gifter, all possible state transitions can be grouped into four categories. ¹⁵

1. Own-action events

When the focal gifter takes an action, she directly affects her own score, the distribution of scores, and potentially her leaderboard rank.

- **Gifting by the focal gifter** ($ev = gift_f(s')$). Gifting increases the focal gifter's score from s to s' , updates the distribution of scores accordingly, and may improve her leaderboard rank depending on the score distribution of other incumbents. Rank updates follow TikTok's deterministic leaderboard rule.
- **Exit by the focal gifter** ($ev = exit_f$). Upon exit, the focal gifter leaves the session, her score is removed from the score distribution, and she becomes idle. Her rank becomes irrelevant after her exit.

2. Other incumbents' actions

Actions taken by other incumbents affect the focal gifter only through changes in the score distribution and relative rankings.

- **Gifting by another incumbent** ($ev = gift_o(s^{m'})$). When another incumbent m increases her score from s^m to $s^{m'}$, the focal gifter's rank is updated only if the other player overtakes her score. Moreover, it changes the distribution of scores.
- **Exit by another incumbent** ($ev = exit_o(m)$). The exit of another incumbent m changes the score distribution and may improve the focal gifter's rank if the exiting incumbent was previously ranked above her.

3. Population flow

Entry and re-entry affect the competitive environment but not the focal gifter's own score. We refer to entry and re-entry as population flow events because they change the set of active competitors in the session without being driven by the focal gifter's strategic choices.

- **Entry** ($ev = entry$). New entrants enter the session with zero score, expanding the score distribution without affecting the focal gifter's rank.

¹⁵Appendix E.1 provides the full formal definition of the transition function, including the exact updating rules for the score distribution and rank assignments.

- **Re-entry** ($ev = re-entry(s^m)$). Re-entering idle gifter m restores her prior score s^m on the leaderboard, potentially bumping down the focal gifter’s rank in the same manner as an opponent’s gifting action.

4. Exogenous state changes

Finally, some state variables evolve independently of strategic actions.

- **Content quality** ($ev = quality(l')$). Content quality changes from l to l' following an exogenous Markov process described in Equation (3).
- **Latent state** ($ev = latent$). The latent variable switches on exogenously, indicating that the session has entered the end stage.

For computational tractability, we simplify how incumbent gifters form expectations about the scores of idle gifters who may re-enter the session. In the model, an idle gifter re-enters with the same score she had at the time of exit. A fully rational gifter would therefore need to keep track of the entire distribution of scores among idle gifters in order to forecast the impact of re-entry on future competition. This distribution can become high-dimensional when many gifters have exited, rendering the model computationally intractable.

Beyond computational concerns, such belief formation is also behaviorally implausible. The leaderboard does not track inactive participants, so gifters would have no direct visibility into the scores of those who have exited the session. Forming fully rational expectations would therefore require gifters to continuously infer and track a detailed distribution of unobserved scores for inactive participants, which is cognitively demanding in a fast-moving livestreaming environment.

To simplify expectations, we assume that gifters hold a non-state-dependent belief about the score of a re-entering idle gifter, denoted by $f(\cdot)$. In estimation, we set $f(\cdot)$ equal to the empirical distribution of scores observed upon re-entry across all sessions in the data.¹⁶ This assumption is analogous to the notion of an oblivious equilibrium (Benkard, Jeziorski, and Weintraub, 2015), in which agents optimize based on stationary aggregate distributions rather than tracking the full high-dimensional state of the market.

Before presenting the Bellman equation, we clarify how gifters form expectations about entry and re-entry. Because the maximum number of active gifters in a session is capped at N , no further entry or re-entry can occur once the number of incumbents reaches this limit. We assume that gifters correctly account for this constraint when forming expectations.

Let $N^{incumbent}$ denote the number of active gifters currently in the session. When $N^{incumbent} < N$, gifters expect two distinct sources of inflow. First, a new entrant may

¹⁶In counterfactual simulations, we enforce belief consistency by iterating between simulated outcomes and agents’ optimal policies until beliefs converge.

arrive at a rate λ_e , in which case at most one new gifter enters the session. Second, idle gifters may re-enter the session. Each idle gifter re-enters independently at rate λ_{re} , implying that the aggregate re-entry rate equals $\lambda_{re} \cdot N^{idle}$. When $N^{incumbent} = N$, no further entry or re-entry can occur.

8.1 Bellman Equation

This section characterizes the forward-looking decision problem faced by an incumbent gifter within a livestreaming session. An incumbent's value, denoted as V_k , reflects not only the immediate utility from current participation, but also expectations about how future gifting, exit, entry, re-entry, and content-quality shocks shape the evolution of competition on the leaderboard.

The Bellman equation below formalizes this continuation value in continuous time. Following [Arcidiacono et al. \(2016\)](#), the value function averages over optimal individual decisions and the stochastic arrival of competitive and environmental events.¹⁷

$$\begin{aligned}
V_k = & \frac{1}{\rho + \lambda_a \cdot N^{incumbent} + (N^{incumbent} < N) \cdot (\lambda_e + \lambda_{re} \cdot N^{idle}) + \sum_{l' \neq l} q_{ll'} + (1 - h) \cdot q_{latent} + h \cdot q_{live}} \times \\
& \left[\underbrace{u^{flow}}_{\text{flow utility}} + \underbrace{\lambda_a \cdot E \max_{s' \geq s, exit} [u^{inst} + V_{k'}(k, gift_f(s')) + \epsilon_{i, s' - s} + V_{k'}(k, exit_f) + \epsilon_{i, e}]}_{\text{focal incumbent } i \text{'s gift \& exit decision}} \right. \\
& + \underbrace{\sum_{m \in N^{incumbent}/i} \lambda_a \cdot \left(\sum_{s^{m'} \geq s^m} \zeta_{m, k, s^{m'}} \cdot V_{k'}(k, gift_o(s^{m'})) \right)}_{\text{other incumbent } m \text{'s gift event}} \\
& + \underbrace{\sum_{m \in N^{incumbent}/i} \lambda_a \cdot \left(1 - \sum_{s^{m'} \geq s^m} \zeta_{m, k, s^{m'}} \right) \cdot V_{k'}(k, exit_o(m))}_{\text{other incumbent } m \text{'s exit event}} \\
& + \underbrace{\lambda_e \cdot (N^{incumbent} < N) \cdot V_{k'}(k; entry)}_{\text{entry event}} + \underbrace{\lambda_{re} \cdot N^{idle} \cdot (N^{incumbent} < N) \cdot \left(\sum_{s^m} f(s^m) \cdot V_{k'}(k; re-entry(s^m)) \right)}_{\text{re-entry event}} \\
& \left. + \underbrace{\sum_{l' \neq l} q_{ll'} V_{k'}(k; quality(l'))}_{\text{Changes in Content Quality}} + \underbrace{(1 - h) \cdot q_{latent} \cdot V_{k'}(k, latent) + h \cdot q_{live} \cdot 0}_{\text{session end}} \right]
\end{aligned} \tag{6}$$

Equation (6) defines the value function for an incumbent gifter in continuous time. The

¹⁷For brevity, we refer the reader to [Arcidiacono et al. \(2016\)](#) for details on the derivation.

value consists of three components. First, while in the session, the incumbent derives flow utility from her leaderboard rank, which creates incentives to gift to improve or defend rank positions. Second, at stochastic decision points, the incumbent optimally decides whether and how much to gift or to exit, taking into account both instantaneous utility and the effects of current actions on future competitive positions. Third, the competitive environment evolves stochastically due to other incumbents' gifting and exit decisions,¹⁸ population flows through entry and re-entry, exogenous changes in content quality, and the possibility that the session ends. Together, these forces determine the incumbent's forward-looking trade-off between immediate gifting costs and future competitive benefits, including the incentive to gift early in order to shape subsequent leaderboard positions.

Given the Type-I extreme value assumption on choice-specific shocks, the incumbent's optimal gifting and exit decisions admit closed-form conditional choice probabilities. At any state k , the incumbent compares the value of exiting the session to the value of gifting different amounts. Each gifting option trades off instantaneous utility—capturing appreciation motives and gifting costs—against the continuation value associated with the resulting leaderboard position. The logit structure implies that actions yielding higher continuation values or higher instantaneous utility are chosen with higher probability, while exit becomes more likely when remaining in the session offers little future competitive benefit.

$$P(s'|k) = \frac{\exp(u^{inst}(s', s, l) + V_{k'(k;gift_f(s'))})}{\left[\sum_{s'' \geq s} \exp(u^{inst}(s'', s, l) + V_{k'(k;gift_f(s'))})\right] + \exp(V_{k'(k;exit_f)})}$$

and the conditional choice probability of exit decisions as

$$P(exit|k) = \frac{\exp(V_{k'(k;exit_f)})}{\left[\sum_{s'' \geq s} \exp(u^{inst}(s'', s, l) + V_{k'(k;gift_f(s'))})\right] + \exp(V_{k'(k;exit_f)})}$$

This policy structure makes clear how stagnation arises in the model. When the continuation value $V_{k'}$ becomes relatively insensitive to additional gifting, such as when score gaps are large or ranks are effectively locked in, the marginal benefit of gifting declines, reducing the probability of further contributions.

9 Estimation and Identification

This section describes how the model parameters are estimated and identified. The estimation strategy exploits the fact that several components of the model (such as entry, re-entry, and content quality transitions) can be estimated outside the full dynamic structure, while the remaining preference parameters are determined by dynamic gifting and exit behavior.

¹⁸ $\zeta_{m,k,s^{m'}}$ denotes the focal incumbent i 's belief about another incumbent m 's gifting action from score s^m to $s^{m'}$ in state k , consistent with equilibrium strategies.

The complete set of parameters includes: (i) parameters governing participation and environmental dynamics, including entry and re-entry arrival rates (θ^e, θ^{re}) , the content quality transition matrix Q_q , and the session termination process (q_{latent}, q_{live}) ; (ii) parameters governing decision timing, including the arrival rate of decision opportunities λ_a ; and (iii) preference parameters, including the price coefficient β_p , appreciation motives (γ_0, γ_1) , rank-based utility $(u_r, r = 1, 3)$, and flow utility from content consumption (γ_2, γ_3) .

9.1 Estimation

We estimate the model using a nested fixed-point (NFXP) algorithm combined with the Method of Simulated Moments (MSM). The estimation proceeds in two stages. In the first stage, we estimate parameters governing entry, re-entry, and content quality transitions $(\theta^e, \theta^{re}, Q_q)$ using maximum likelihood. These components do not depend on strategic gifting incentives and therefore can be identified without solving the full dynamic game.

In the second stage, we estimate the remaining structural parameters by minimizing an MSM objective function that matches model-implied moments to their empirical counterparts. The estimation follows the standard NFXP structure. In the inner loop, fix estimated primitives $(\hat{\theta}^e, \hat{\theta}^{re}, \hat{Q}_q)$ from the first stage, we solve the equilibrium policy for each candidate vector of parameters $\theta = (\lambda_a, \beta_p, \gamma_0, \gamma_1, u_1, u_3, \gamma_2, \gamma_3, q_{latent}, q_{live})$. To solve for equilibrium, we use the Bellman equation (6) to iterate on the value function until convergence (at precision $1e - 5$), and recover the gifters' optimal policy (on exit, gift, and doing nothing) using the value function. Then, we forward simulate the entire environment—including the gifters' actions, entry, reentry, content quality evolution, and session termination—for 20,000 sessions to generate model-implied moments. In the outer loop, we compare the model-implied moments, generated from the equilibrium policy functions, with their empirical counterparts observed in the data. We compute the following MSM objective function:

$$\Pi(\theta) = g(\theta)'Wg(\theta) \quad (7)$$

where $g(\theta)$ is the vector of moment conditions (differences between empirical and model-implied moments), and W is the weighting matrix. The second column of Table 5 provides a comprehensive list of target moments we match. We search for the optimal parameter vector θ that minimizes MSM objective value:

$$\hat{\theta} = \arg \min_{\theta} \Pi(\theta) \quad (8)$$

This minimization is carried out using NelderMead optimizer built in Julia, where the convergence precision is set to be $1e - 5$. The third column of Table 5 provides the model-implied moments after optimization.

Table 5: Moments for MSM Estimation

Targeted moment	Empirical moment	Model-implied moment
<i>Session-level Moments</i>		
total gifts	2692	2797
total gifts% at high-quality content	0.800	0.76
exit probability at high-quality content	0.112	0.137
exit probability at low-quality content	0.127	0.162
session duration	128	134
<i>Re-entry Event Moments (6-min post v.s. 6-min pre)</i>		
change in gifts for bumped top 1	25.2	22.22
change in gifts for bumped top 3	15.5	17.3
<i>Conditional Choice Probability</i>		
p_{12}	0.719	0.792
p_{13}	0.186	0.185
p_{14}	0.053	0.021
p_{15}	0.037	0.000
p_{16}	0.002	0.000
p_{23}	0.804	0.877
p_{24}	0.155	0.120
p_{25}	0.040	0.002
p_{26}	0.000	0.000
p_{34}	0.855	0.924
p_{35}	0.121	0.075
p_{36}	0.022	0.000
<i>Gifting Dynamics</i>		
total gift% at stage 1	0.290	0.323
total gift% at stage 2	0.277	0.291
total gift% at stage 3	0.254	0.247
total gift% at stage 4	0.176	0.139

Note: This table provides a complete list of moments used for MSM estimation. The columns from left to right are the name of the target moments, the empirical moments calculated from the observational data, and the model-implied moments from the data simulation. At the session level, we calculate the total gifts, the percentage of gifts for high-quality content, the exit probability for high-quality content, and the exit probability for low-quality content. Leveraging re-entry events, we calculate the gift increment by taking the differences. We separately calculate the gift increment for gifters bumped from rank 1 and from rank 3. Conditional choice probability $p_{s,s'}$ is the probability of viewers' gifts to s' conditional on them being originally at s and sending a positive amount of gifts. For gifting dynamics, we divide the sessions into four equal-length stages and calculate the gift percentage contributed in each stage.

9.2 Identification

This section discusses parameter identification. Table 6 summarizes the data moments we used to identify our model parameters.

Table 6: Summary of Data Moments Used to Identify Each Parameter.

Concept	Parameter	Data Moments
Rank utility for top 1	u_1	The increment of gift after No. 1 gifters get bumped
Rank utility for top 2, 3	u_3	The increment of gift after No. 3 gifters get bumped
Appreciation motive	γ_0, γ_1	Gifting at varying levels of content quality
Flow utility for content consumption	γ_2, γ_3	Exit frequency at varying levels of content quality
Price coefficient	β_p	Choice probability conditional on gifting
Arrival rate of decision making	λ_a	Exclusion restriction of rank on utility
Arrival rate of entry	λ_e	Length of time before entry
Arrival rate of re-entry	λ_{re}	Length of time before re-entry
Content quality transition rate	$q_{l,l'}$	Length of time between different content quality levels
Arrival rates of session ending	q_{latent}, q_{live}	Session duration and gifting dynamics

Competitive Motive The competitive motive is identified from gifters’ behavioral responses to exogenous changes in leaderboard rankings induced by re-entry events. As documented in Table 2, when a re-entering gifter bumps an incumbent down from a top rank, the bumped gifter increases her gifting immediately after the event. Because re-entry shocks mechanically alter relative ranks without directly affecting content quality or prices, the resulting increase in gifting reflects a revealed willingness to pay to regain a higher leaderboard position.

To quantify this response, we isolate a six-minute window around each re-entry event and compute the change in gifting between pre- and post-event periods. We construct separate moments for gifters bumped from the No. 1 and No. 3 positions. These rank-specific gifting responses identify the rank-based utility parameters by mapping observed monetary responses to the marginal value of occupying higher leaderboard positions.

Appreciation Motive and Price Coefficient For gifters who rank below the top five, gifting does not affect leaderboard positions and therefore reflects a trade-off between appreciation utility and monetary cost. Variation in their gifting behavior identifies the appreciation motive and price sensitivity.

Conditional on making a positive gift, the distribution of gifting levels identifies the baseline appreciation parameter and the price coefficient. Higher observed gifting levels indicate either stronger appreciation or lower price sensitivity. These two forces are separately identified by exploiting functional form restrictions: appreciation utility is concave in gifting amount, whereas monetary cost enters linearly. As a result, the curvature of the choice probabilities over gifting levels identifies the price coefficient β_p and baseline appreciation γ_0 .

Moreover, changes in gifting behavior across different levels of content quality identify

the quality-dependent appreciation parameter, i.e., γ_1 . Specifically, we target the share of gifts contributed during high-quality content periods, which isolates the extent to which appreciation responds to content quality rather than prices.

Flow Utility from Content Consumption Utility from consuming content changes gifters’ consumption value from staying in the session. Therefore, a higher content consumption value reduces the gifters’ exit probability. Moreover, the different levels of exit frequency at varying levels of content quality identify gifters’ baseline utility γ_2 and marginal utility from content consumption γ_3 .

Gifting Arrival Rate In addition, the gifting arrival rate is jointly identified using both conditional gifting behavior and aggregate gifting intensity. Conditional choice probabilities pin down how much to gift for each decision opportunity, while total gifts per session reflect the frequency with which such opportunities arise. Holding conditional gifting behavior fixed, a lower arrival rate mechanically implies fewer gifts at the session level. Therefore, matching both conditional gifting probabilities and total gifts per session allows the model to separately identify the arrival rate of decision opportunities from utility parameters.

Arrival rates of session ending The session-ending process consists of two sequential stages: a latent transition into the terminal phase and the final termination of the session. These two stages are governed by the arrival rates q_{latent} and q_{live} , respectively. The parameter q_{latent} determines how long a session remains in the regular stage before entering the terminal phase, while q_{live} governs how quickly the session ends once the terminal phase begins.

These two arrival rates are jointly identified from session duration and gifting dynamics. Conditional on average session length, there is a trade-off between the timing of the latent switch and the duration of the terminal phase: sessions that enter the terminal phase later tend to have a shorter terminal phase before ending. A smaller q_{latent} implies a longer regular stage, whereas a larger q_{live} implies a rapid termination once the session enters the terminal phase.

This structure generates distinct patterns in gifting dynamics. In particular, when the terminal phase is short, gifting exhibits a sharp decline near the end of the session, reflecting viewers’ anticipation of imminent termination.

Other Arrival Rates. The arrival processes for other events, such as entry, re-entry, and content-quality transitions, are assumed to follow exponential distributions. Their arrival rates are identified from the empirical frequency of these events in the data: higher arrival rates correspond to more frequent occurrences.

9.3 Estimation Results

In this section, we focus on the estimated preference parameters that govern viewers’ gifting motives and competitive behavior. Parameters governing participation timing and environmental dynamics, such as entry, re-entry, content quality transitions, decision chance, and session termination, are reported and discussed in Appendix E.4.

Table 7 presents the model estimates for all parameters. The cost coefficient of gifting is significant at 1.561, which provides an interpretation about the monetary worth of utilities. Because gifters generate 1.561 units of disutility whenever they spend one hundred TikTok coins on gifts, and each 100 coin is worth \$1.4. Therefore, one unit of utility is worth $(1/1.561) * 1.4 = 0.896$ dollar.

With the above conversion rule between utility and monetary value, I could directly interpret gifters’ willingness to gift. Competitive motive stems from the flow utility of ranking the top positions. According to the estimates, utility for top 1 position is positive and significant at 0.737 and the utility for top 3 position is positive and significant at 0.227. Our estimates also validate that gifters value higher positions to lower positions, i.e., $u_1 > u_2 = u_3$. We further interpret the rank-based utility by its monetary value. Gifters value each minute of staying in the top 1 positions at $0.737 * 0.896 = 0.660$ dollar. Similarly, each minute of staying in the top 2 or 3 positions is worth $0.227 * 0.896 = 0.203$ dollar.

For the appreciation motive, gifters’ baseline appreciation factor and marginal appreciation factor are both positively significant at 1.374 and 0.689. The coefficient γ_0 captures baseline appreciation utility from gifting, while γ_1 measures how this appreciation motive is amplified during high-quality moments. Content quality enters the utility function through a discretized state $l_t \in \{l^{low}, l^{high}\}$, where l^{high} corresponds to minutes with substantially above-median audience engagement. Since the utility from showing appreciation is concave in the gifting amount, we interpret the appreciation motives through considering how much is the utility of showing appreciation when a gifter contributed one-hundred TikTok coins (worth \$1.4). Firstly, baseline appreciation factor is positive significant at 1.374, indicating how much gifters value showing appreciation when the content quality is of median level. We find that gifters generate 0.853 dollar worth of utility from showing appreciation at a median-quality content.¹⁹ Second, appreciation factor is positive and significant, indicating utility from showing appreciation is greater with higher content quality. For the same one-hundred coins spend, if the content quality is improved by one standard-deviation, the value of showing appreciation will increase by 0.428 dollar.²⁰

¹⁹The interpretation of appreciation motive is slightly difficult given the utility from showing appreciation is concave in gifting amount. One-hundred coins generates $1.374 * \ln(2) = 0.952$ unit utility, which is worth $0.952 * 0.896 = 0.853$ dollar.

²⁰Note the numerical values of l^{low} and l^{high} are constructed by normalizing minute-level likes relative to the session median, one-standard deviation of content quality improvement further improves the utility from showing appreciation by $0.689 * \ln(2) = 0.477$, which is worth $0.477 * 0.896 = 0.428$ dollar.

Table 7: Model Estimates

Concept	Parameter	Estimates
Competition motive		
Rank utility for rank 1	u_1	0.737 (0.012)
Rank utility for rank 2/3	u_3	0.227 (0.001)
Appreciation motive		
Baseline appreciation	γ_0	1.374 (0.0004)
Quality multiplier	γ_1	0.689 (0.001)
Content consumption		
Baseline value	γ_2	0.130 (0.026)
Quality multiplier	γ_3	0.100 (0.012)
Cost function		
price coefficient	β_p	1.561 (0.016)

Lastly, we find that viewers derive substantial consumption value simply from watching the livestream itself. The baseline consumption value is significant at 0.13, indicating viewers value watching at $0.130 * 0.896 = 0.116$ dollar per minute at median content quality. Moreover, the return of quality from watching content is 0.1, meaning that a one-standard-deviation increase in content quality further enhances this content consumption value by $0.1 * 0.896 = 0.089$ dollar per minute.

Our model estimates validate and improve our understanding of viewers' gifting motive. We find gifters' competitive motive is not only significant, but also substantial in its monetary value. They place a greater value on higher rankings, i.e., $u_1 > u_2 = u_3$. Moreover, we also find a substantial size of appreciation motive for gifting and consumption value for watching and participating in the session.

10 Counterfactuals

In this section, we conduct counterfactual analyses. The first is a study of the economic value of the leaderboard (Section 10.1). In the second counterfactual (Section 10.2), we evaluate

an alternative leaderboard design of discounted scoring system to manage stagnation and sustain competition.

10.1 Economic Value of the Leaderboard

In our first counterfactual, we evaluate the economic value of the leaderboard by setting the rank-based utility to zero and measuring the resulting revenue loss. As shown in Table 8, the average total revenue per session is \$39, with leaderboards contributing 43% of this amount and generating \$0.154 per minute. This highlights the substantial role leaderboards play in driving revenue on TikTok LIVE.

To explore heterogeneity, we divide sessions into long and short ones and find that leaderboards are relatively less important in long sessions: both the share of leaderboard-driven revenue and the revenue per minute attributable to the leaderboard are smaller. The leaderboard contributes to 40% of total revenue and generates \$0.150 per minute in long sessions, whereas the leaderboard contributes to 46% of total revenue and generates \$0.162 per minute in short sessions.

The reason for the above finding is that long sessions are more prone to the negative effects of stagnation. Early in a session, gifting dynamics look similar across both types, but as the session progresses, top contributors accumulate a large competitive advantage. Once this happens, other viewers perceive little chance of improving their rank, causing the leaderboard to “freeze up.” Because long sessions extend further into this stagnation phase, they experience a larger proportion of time in which leaderboard competition no longer generates incremental revenue. We provide supporting evidence for this intuition using a proxy for stagnation: the inverse of the turnover rate in each session. Specifically, we define the turnover rate as the number of times the top three gifters are displaced on the leaderboard due to others surpassing them through gifting, normalized by session duration (minutes). We find that turnover rates are lower in long sessions (0.05) than in short ones (0.057), confirming that long sessions suffer more severely from stagnation.

Table 8: How Much Revenue is Contributed by the Leaderboard

average total revenue per session: \$39			
	revenue from leaderboard%	revenue per minute from leaderboards	turnover rate
average session (108 min)	43%	\$0.154	
long session (140 min)	40%	\$0.150	0.050
short session (80 min)	46%	\$0.162	0.057

10.2 Score Design

In the second counterfactual, we study the optimal score aggregation rule—how gifting contributions are mapped into leaderboard scores. Our empirical analysis documents a stagnation pattern: a negative relationship between score disparity and players’ willingness to gift. As score gaps widen, lower-ranked gifters become discouraged from competing, since closing the gap becomes increasingly costly and unlikely to be rewarded (Table 3). The objective of this counterfactual is therefore to design a time-discounted score rule that mitigates stagnation and sustains competitive engagement.

Before introducing alternative score designs, it is useful to clarify why stagnation in livestreaming differs from that in standard tournament settings. Unlike conventional tournaments, where rewards are realized only at the end, leaderboard positions in livestreaming generate flow utility throughout the session. This reward timing encourages participants to exert effort early to secure top positions, while reducing trailing players’ incentives to close gaps that may not pay off before the session ends. Moreover, stagnation cannot be mitigated by reducing performance feedback, as flow utility relies on continuous leaderboard visibility. These features motivate score design policies that manage stagnation without suppressing real-time ranking information.

To address this problem, we propose a discounted score design in which gifters’ accumulated scores depreciate over time. This design places greater weight on recent contributions and limits the persistence of early advantages. Operationally, we model discounting as a stochastic process: discount events arrive according to an exponential distribution with rate $\lambda = 1/30$, implying that scores are discounted once every 30 minutes on average. When a discount event occurs, a gifter’s score is multiplied by a discount factor β and rounded to the nearest point on the score grid.²¹ When $\beta = 1$, scores are never discounted and rankings reflect cumulative contributions, corresponding to the status quo on TikTok LIVE. Smaller values of β imply stronger discounting, with earlier contributions carrying less weight over time.

Discounting has two opposing effects on revenue. If β is too large (weak discounting), scores still accumulate and eventually stagnant, with top contributors pulling far ahead and discouraging further competition. If β is too small (strong discounting), forward-looking gifters anticipate that their contributions will lose value too quickly and may withhold gifts altogether.

Based on this trade-off, Table 9 varies the score discount factor β and identifies an optimal value of $\beta = 0.55$, which maximizes platform revenue and generates a 19.7% increase in total gifts relative to the baseline leaderboard.²² Consistent with the model’s prediction, more intensive score discounting (lower β) reduces stagnation, as reflected in higher turnover

²¹Recall that the score grid in the model is $[0, 100, 300, 600, 1000, 1500]$.

²²A full characterization of outcomes across β in increments of 0.05 is reported in Appendix D.1.

Table 9: Counterfactual Outcomes under Alternative Score Discounting

β	total gifts (\$)	% change in revenue	turnover rate	number of gifters
1.00	39.16	0.0%	0.054	7.10
0.90	39.16	0.0%	0.054	7.10
0.80	39.16	0.0%	0.054	7.10
0.70	43.65	11.5%	0.086	7.17
0.60	46.79	19.5%	0.129	7.50
0.55	46.87	19.7%	0.129	7.50
0.50	46.67	19.2%	0.130	7.49
0.40	46.26	18.1%	0.157	7.73
0.30	46.09	17.7%	0.159	7.74
0.20	45.77	16.9%	0.159	7.76
0.10	43.98	12.3%	0.167	7.78

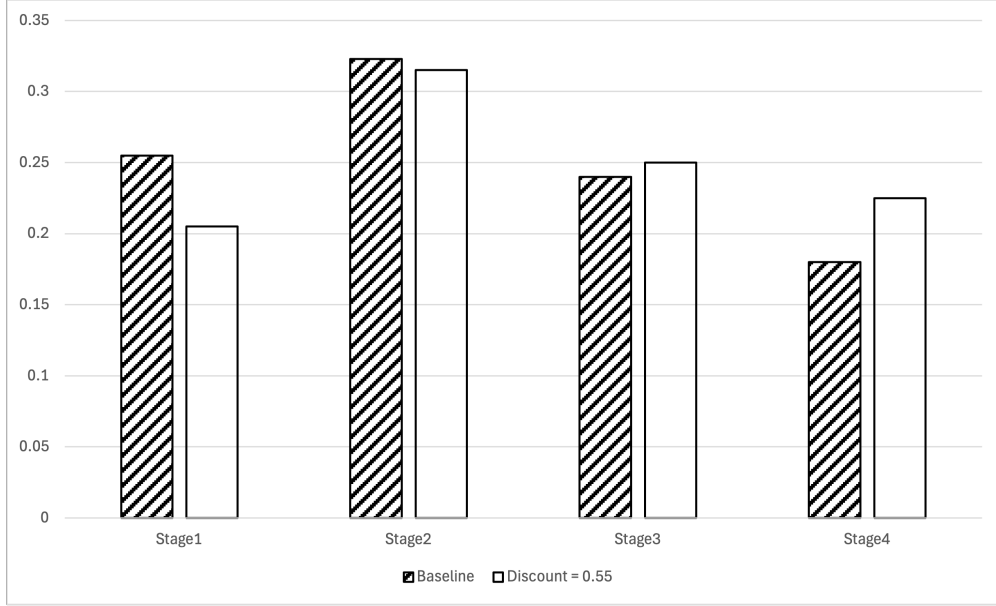
among the top-three gifters. Importantly, reducing stagnation is not always optimal: at the revenue-maximizing discount factor, the implied turnover rate is 0.129, meaning that the identity of at least one top-three gifter changes with a 12.9% probability per minute. Furthermore, stronger discounting increases participation in gifting. As β decreases, more viewers choose to gift a positive amount, reflecting a greater perceived chance of improving rank. The optimal discount factor therefore balances increased participation and stronger incentives for late-stage gifting against diminished incentives for early gifting.

The effectiveness of discounted scoring in managing stagnation depends on forward-looking gifters reallocating their contributions over the course of a session. Under discounted scoring, gifters delay part of their contributions to later stages rather than concentrating effort at the beginning. Figure 6 illustrates this shift for the largest contributor in each session.²³ The horizontal axis represents session progression, divided evenly into four stages, while the vertical axis shows the share of total gifts contributed at each stage. Relative to the baseline without discounting, top gifters reduce their spending in early stages and increase it in later stages. By slowing the accumulation of early advantages, discounted scoring mitigates stagnation and sustains competition throughout the session.

Table 10 further explores heterogeneity by session duration and popularity, examining how the optimal discount factor varies with these characteristics. I first explore the heterogeneity in session duration while holding popularity constant. The optimal discount factor is lower in long sessions ($\beta = 0.55$) than in short sessions ($\beta = 0.6$). The intuition is that in

²³The pattern is the same for the second and third largest contributors in the session.

Figure 6: Gifting Dynamics Baseline V.S. Discounted Scores



Note: The horizontal axis represents session progression, with the session split equally into four stages. The vertical axis measures the percentage of total gifts that No.1 gifters contributed in each stage. The dashed bar is the baseline gifting dynamics without discounting. And the white bar is the gifting dynamics with optimal discounting. We find that under discounted scoring, gifting is delayed to the later stage of the session. The percentage of gifts is low at stage 1 because only a small number of gifters have entered at stage 1.

Table 10: Heterogeneity in Optimal Score Discounting

	discount factor β	% change in revenue	session duration (min)	session popularity	turnover rate
average session	0.55	+19.7%	110	0.65	0.054
heterogeneity 1: session duration					
long session	0.55	+23.5%	140	0.65	0.050
short session	0.60	+13.3%	80	0.65	0.057
heterogeneity 2: session popularity					
more popular session	0.55	+19.5%	110	0.77	0.054
less popular session	0.55	+20.2%	110	0.54	0.054

longer sessions, early advantages can persist for a greater duration, potentially amplifying stagnation and weakening competition in later stages. To counteract this, more aggressive discounting (a smaller β) is needed to prevent the top gifters from pulling far ahead in long sessions. Moreover, because stagnation is more severe in long sessions, these sessions benefit more from discounting, as reflected in the larger percentage improvement in total revenue (+23.5%) in long sessions than in short sessions (+13.3%). Consistent with this, we find that the turnover rates are lower in long sessions (0.050) than in short sessions (0.057). For the second heterogeneity analysis, we then compare more popular and less popular sessions while

holding the average duration fixed at 110 minutes. In this case, we do not find significant differences either in the optimal discount factor or in the revenue gains from discounting. This suggests that session popularity does not meaningfully moderate the stagnation level. Consistent with this, we find that the turnover rates are similar between more and less popular sessions.

From a managerial perspective, these results highlight that leaderboard design is not one-size-fits-all. The revenue-maximizing level of score discounting depends critically on how long competitive interactions are expected to last. In sessions that are likely to run longer, early gifting can entrench dominant positions for an extended period, making stagnation more severe and reducing overall participation. In such environments, more aggressive score discounting helps sustain competition by periodically resetting effective advantages. In shorter sessions, by contrast, excessive discounting can undermine incentives to gift early, leading to smaller gains. These findings suggest that platforms can improve performance by tailoring leaderboard rules to the anticipated duration of competitive interactions rather than applying a uniform design across all sessions.

10.3 Discussion of Counterfactual Assumptions

The counterfactual analyses rely on several behavioral and equilibrium assumptions regarding how viewers and content creators respond to alternative leaderboard designs. Appendix D presents a set of auxiliary empirical exercises that assess the plausibility of these assumptions by examining environments in which they may be violated.

First, we examine whether alternative leaderboard designs induce changes in content production by creators. In the counterfactual analyses, we hold the content-production process fixed and focus on how leaderboard design affects demand-side gifting behavior. Empirically, we find no systematic changes in content quality or session duration following leaderboard-induced ranking changes, supporting the assumption that content production is approximately invariant to the scoring rules considered.

Second, in counterfactual simulations, we hold entry and re-entry processes fixed. This implicitly assumes that the relationship between participation and its determinants, such as day-of-week and session popularity, remains invariant to changes in leaderboard design. This assumption is plausible in our setting for two reasons. First, day-of-week captures routine-driven availability and responsiveness to live-stream notification, which are orthogonal to leaderboard rules. Second, session popularity, proxied by real-time likes, affects participation through exposure and recommendation channels rather than through competitive incentives. Consistent with our reduced-form evidence, we do not find that participation patterns are systematically conditioned on within-session competition, supporting the stability of these relationships under alternative leaderboard designs. Moreover, we find no

evidence that alternative score designs significantly alter content production, as measured by content quality and session duration, further supporting the invariance of participation processes in our counterfactual analysis.

Third, we investigate whether discounted scoring rules alter viewers’ underlying preferences for leaderboard ranks. While discounted scores could, in principle, affect how viewers perceive the value of leaderboard positions, we find limited empirical evidence that viewers’ rank preferences materially change under discounting. This suggests that discounted scoring primarily affects competitive dynamics through incentives rather than through shifts in preferences.

Fourth, we explore whether discounted scoring rules generate frustration responses that could undermine their effectiveness. Although frustration is not directly observable in the data, the behavioral adjustments implied by the counterfactual suggest that viewers can adapt to discounted scoring in ways that mitigate frustration. First, gifting is redistributed from earlier to later stages of the session, indicating that viewers are less likely to concentrate spending early and subsequently experience large score discounts. Second, discounted scoring increases equality in viewers’ opportunities to compete for top positions, which the platform can leverage to frame the rule as enhancing fairness and participation. Consistent with this interpretation, we find greater participation in gifting and higher turnover among top-ranked positions under discounted scoring. Taken together, these findings suggest that the counterfactual revenue gains are unlikely to be driven by behavioral backlash under the scoring rules we consider.

Finally, our heterogeneity analysis is conducted based on realized session duration. This measure should be interpreted as a noisy proxy for creators’ planned session length. An implicit assumption is that platforms can anticipate session duration when implementing duration-specific leaderboard designs. Industry evidence suggests that established creators typically predetermine the structure and approximate duration of their livestreams before going live. Platforms can observe or obtain information about planned duration *ex ante* through coordination with creators, whereas viewers do not observe this plan and therefore face uncertainty about the exact timing of session termination. Accordingly, the model’s stochastic session-ending process captures viewers’ subjective uncertainty, while the heterogeneity analysis conditions on expected session length proxied by realized duration (see Appendix D.6 for industry evidence).

11 Conclusion and Discussion

This paper studies how leaderboard design shapes user engagement and platform revenue on digital platforms. Using data from TikTok LIVE, the reduced-form evidence in Section 5 shows that leaderboards fundamentally reshape viewers’ gifting behavior by introducing

competition among them, on top of an intrinsic appreciation motive for high-quality content. Importantly, we document a challenge of stagnation faced by leaderboards when cumulative performance metrics are used to rank viewers: when top viewers cumulate their competitive advantage and are far ahead on the leaderboard, this discourages viewers’ effort to compete, because they perceive little opportunity to change their relative standings. This challenge presents an opportunity for TikTok LIVE to improve revenue by redesigning its leaderboard scoring rules.

To quantify gifting motives, assess the economic impact of leaderboards, and evaluate alternative scoring designs, Section 6 develops and estimates a dynamic model of viewer competition over time. Our counterfactual analysis quantifies the economic significance of leaderboards in driving platform revenue, revealing that competitive incentives account for nearly half of total platform revenue. Heterogeneous analysis continues to find that the effectiveness of leaderboards in long sessions is substantially constrained by stagnation problems later in the session. Motivated by this, we evaluate discounted scoring systems that place greater weight on recent gifting contributions and less weight on more distant ones in ranking viewers. The optimal scoring system balances initial participation and the competitive motive at a later stage, leading to a substantial improvement in platform revenue. More broadly, this paper contributes to the leaderboard literature by shifting the focus from evaluating the effects of leaderboards to designing scoring rules that shape user incentives and platform outcomes.

The findings presented in this paper have general implications for digital platforms that feature real-time leaderboards during consumption sessions. In particular, this paper provides systematic evidence that leaderboards granting users social utility from high relative performance during consumption are effective in motivating user engagement. This class of leaderboard designs is widely adopted across digital platforms, yet has received limited economic evaluation. By quantifying its effects on behavior and revenue, this paper helps validate the role of social-utility-based leaderboards as an important design tool for digital platforms.

These insights have direct relevance for livestreaming platforms, a rapidly growing market valued at 88 billion in 2024 and projected to reach 600 billion by 2033 (Imarc, 2025). On leading livestreaming platforms such as Twitch, TikTok LIVE, Bigo Live, and Kuaishou, voluntary gifting constitutes the primary source of revenue. By identifying viewers’ competitive and appreciation motives and evaluating leaderboard designs, this paper provides guidance on how platforms can more effectively harness leaderboards to drive revenue. Beyond livestreaming, similar leaderboard designs are widely used in exercise and learning platforms such as Peloton and Kahoot. While these platforms monetize primarily through subscriptions, prior research shows that higher user engagement is predictive of higher subscription retention and lower churn (Pattabhiramaiah, Overby, and Xu, 2022; Godinho de

Matos, Mamadehussene, and Ferreira, 2026). As a result, leaderboards that successfully motivate engagement may indirectly contribute to revenue even when subscriptions are the primary monetization channel.

Finally, the findings highlight the importance of addressing stagnation in leaderboard-based systems. When early leaders accumulate large advantages, users may become discouraged from competing, reducing overall engagement. The results suggest that introducing discounting into leaderboard scoring rules can manage stagnation and sustain competition over time. Anecdotal evidence from platforms such as Peloton and Kahoot, as well as industry commentary, points to widespread user frustration with stagnant leaderboards.²⁴ This paper provides systematic economic evidence supporting the need for leaderboard designs that actively manage stagnation rather than allowing competitive outcomes to become permanently locked in.

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²⁴For example: [Feeling a bit discouraged by our output numbers/Leaderboard](#) and [Kids don’t know how to handle losing](#).

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A Additional Institutional Background and Descriptive Evidence

A.1 Distribution of Topic Categories

Table .11: Session Characteristics by Content Category

Category	#Session	#Creator	#Viewer	Session duration(mins)
1. Music	3775	76	1875	142
2. Daily Life	2661	67	1474	105
3. Chat&Interview	1429	47	2114	126
4. Art	1099	13	3897	126
5. Dance	1057	20	3582	117
6. Food	966	30	6654	121
7. Learn	392	22	2045	101
8. Gaming	338	13	1724	402
9. Beauty&Fashion	227	6	7456	129
10.Pets	155	3	2732	97
11. Education	117	6	5356	147
12. Relationship	114	3	490	125
13. DIY	90	8	2009	183
14. Fitness	56	2	955	163
15. Outdoors	29	5	4587	91
16. Shopping	14	3	2990	251
17. Multi-Guest	6	1	1134	70
18. Fandom	2	1	2637	281

TikTok LIVE supports various category hashtags to help viewers discover content aligned with their interests. Before starting a live session, the content creator can decide whether to include a hashtag in their live stream title, which could enhance visibility and engagement by categorizing their content. On TikTok, hashtags (words or phrases preceded by a "#" symbol) act as searchable keywords that categorize content and make it discoverable by other users. They help the TikTok algorithm understand your video and show it to a relevant audience. Table .11 ranks the popularity of categories by the number of sessions.

I summarize the number of sessions/creators in each category and the average number of viewers/session durations per session in my sample²⁵.

The top-five categories on TikTok LIVE are "Music", "Daily Life", "Chat&Interview", "Art", and "Dance", consisting of 69% of total sessions in my sample. There is a substantial variation in viewer sizes for an average session across different categories. For example, Beauty&Fashion has the highest viewer size per session, with over 7000 viewers attending an average session, while Relationship has the lowest viewer per session(only 490 viewers on an average session). If I rank categories by the viewer size per session, the top five categories are "Beauty&Fashion", "Food", "Education", "Outdoors", and "Art". The session duration for an average session is around two hours across categories, except Gaming, Fandom, and Shopping, which have longer sessions(average session duration is over four hours).

B Additional Robustness Analyses

B.1 Event-Study Evidence around Re-entry of Top-1 and Top-3 Gifters

I conduct an event-study analysis around the re-entry of top gifters to examine whether there are pre-trends in gifting behavior prior to re-entry and to assess the timing of behavioral responses. Specifically, I focus on gifters who were previously ranked in the No. 1 or No. 3 positions but were subsequently bumped out of these positions because the re-entry of gifters at No. 1 or No. 3 positions.

For this analysis, I pool all re-entry events associated with the No. 1 and No. 3 positions and isolate a 13-minute window centered around the re-entry moment t_0 , spanning from 6 minutes before to 6 minutes after re-entry. I estimate the following regression:

$$\text{gift}_{ist} = \sum_{\Delta t=-5}^6 \alpha_{\Delta t} * I\{t = t_0 + \Delta t\} + \lambda * \text{others' like}_{ist} + \psi * \text{audience size}_{st} + \phi * t + \theta_s + \delta_{ic(s)} + \epsilon_{ist}$$

where $I\{t = t_0 + \Delta t\}$ is an indicator for each minute relative to the re-entry moment. The specification takes the minus-6 minute as the baseline, and controls for contemporaneous engagement from other viewers, session-level audience size, linear time trends, session fixed effects, and individual-by-session fixed effects.

Table .12 reports the estimated coefficients $\alpha_{\Delta t}$. The estimates show no statistically significant increase in gifting behavior in the minutes leading up to the re-entry event, indicating the absence of pre-trends. In contrast, I observe a sharp and statistically significant increase

²⁵I exclude 18475 sessions (56% of all sessions) which do not include hashtags in their live stream titles. I also exclude 24 content creators who never used hashtags for their live sessions

in gifting exactly at the moment of re-entry ($\Delta t = 0$), followed by a sustained elevation in gifting over the subsequent minutes.

The absence of pre-trends, together with the precise temporal alignment of the gifting response with the re-entry event at the minute level, suggests that the observed increase in gifting is unlikely to be driven by unobserved gradual changes in viewer engagement or other time-varying confounders. Instead, the results support a causal interpretation in which re-entry into salient leaderboard positions directly triggers intensified competitive gifting behavior of gifters whose positions are involuntarily taken away.

B.2 Placebo Test for Re-entry Events

To further assess whether the re-entry effects are driven by spurious time trends or anticipation, I conduct a placebo test by artificially assigning the re-entry time to be 10 minutes earlier than the actual re-entry moment. I then re-estimate the following specification using this placebo event time:

$$\begin{aligned} \text{gift}_{ist} = & \sum_{k=1}^6 \alpha_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k, t \geq t_0\} + \sum_{k=1}^5 \beta_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k\} \\ & + \lambda * \text{others' like}_{ist} + \psi * \text{audience size}_{st} + \phi * t + \theta_s + \delta_{ic(s)} + \epsilon_{ist} \end{aligned}$$

Table .13 reports the estimated coefficients from the placebo regression. I find that all placebo treatment coefficients $\alpha_{k,k+1}$ are statistically insignificant and economically small. This indicates that, absent an actual re-entry event, bumped gifters do not exhibit any systematic change in gifting behavior.

Together with the event-study evidence, this placebo test supports the interpretation that the observed increase in gifting is causally driven by the re-entry of top-ranked gifters, rather than by underlying trends, anticipation effects, or other time-varying confounders.

C Evidence Informing Model Specification

C.1 Construction of Content Quality States

I discretize minute-level likes into two quality states using a median split. Minutes with likes below (above) the session-level median are classified as low-quality (high-quality).

To assign numerical values to each state, I compute the average deviation of likes from the median within each group and normalize this deviation by the standard deviation of likes. The resulting normalized values define l^{low} and l^{high} , which enter the instantaneous utility function.

Table .12: Event Study of Bumped Viewers' Behavior in Re-entry Event

	Gift by bumped-down viewers
α_{-5}	1.4912 (1.0713)
α_{-4}	0.5712 (1.0487)
α_{-3}	1.8009* (1.0879)
α_{-2}	-0.5842 (0.9314)
α_{-1}	-1.1761 (0.7399)
α_0	3.0330* (1.5651)
α_1	4.5355* (2.5288)
α_2	3.3065* (1.8108)
α_3	3.1855* (1.6426)
α_4	4.1836*** (1.5301)
α_5	3.5492** (1.5374)
α_6	5.2772** (2.3090)
Observations	155,311
Controls ¹	YES
Session FE	YES
Viewer×Creator FE	YES

Note: Standard errors are in parentheses. Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level. ¹ Including total likes by others, audience size, and number of minutes since the start of the session.

Table .13: Placebo test for the re-entry event

	Gift by bumped-down viewers					
	1 to 2	2 to 3	3 to 4	4 to 5	5 to 6	6 to 7
$\alpha_{k,k+1}$	2.7118 (4.2026)	-5.3364 (8.3426)	-0.9555 (1.9643)	0.7900 (0.9740)	-0.2294 (0.7932)	0.6146 (0.7142)
Observations	20,028					
Controls	YES					
Session FE	YES					
Viewer $\times$ Creator FE	YES					

Note: Controls and fixed effects include the total likes sent by others $like_{ist}$, the size of the viewers of the session audience $size_{st}$, the number of minutes since the start of the session t , session fixed effect θ_s , and individual times creator fixed effect $\delta_{ic(s)}$. Standard errors are in parentheses. Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level.

This normalization ensures that the magnitude of quality effects is comparable across sessions with different baseline engagement levels.

C.2 Evidence on Liking as a Proxy for Content Quality

In my model, I use total likes to proxy the content quality. The underlying assumption is that liking activities show people’s attitudes towards content quality. I want to rule out an alternative case in which people send like not just to express their likes for the content.

The alternative explanation is that viewers send likes because there are many people gifting, making the live session very entertaining. The following regression examines

$$\begin{aligned} \text{total likes}_{st} = & \alpha * \text{total gifts}_{st-1} + \beta * \text{total comments}_{st-1} \\ & + \lambda * \text{total likes}_{st-1} + \psi * \text{audience size}_{st} + \phi * \text{since start}_{st} + FE_s \end{aligned}$$

C.3 Individual-Level Liking and Gifting Conditional on Content Quality

In my main analysis, I do not model liking activity at the individual level; instead, I include liking at the aggregate (session-minute) level. The reason is that a “like” primarily reflects viewers’ appreciation of the content, rather than a strategic action that directly affects an individual viewer’s gifting decision. As such, liking activity matters for gifting only insofar as it captures time-varying content quality that is common to all viewers in the session.

To assess whether individual-level liking activity has an independent effect on individual gifting behavior beyond content quality, I estimate the following regression:

Table .14: Spillover in Liking Activities Across Viewers

	total likes
lag total gift	-0.0002 (0.0002)
lag total comment	0.5515*** (0.0678)
lag total like	0.7418*** (0.0025)
Observations	1,236,889
Session FE	YES
Controls ¹	YES

Note: Standard errors are in parentheses. Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level. ¹ Including viewer size, number of minutes since the start of the session.

$$\text{gift}_{ist} = \lambda * \text{like}_{ist} + \theta_{st} + \delta_{ic(s)}$$

where θ_{st} denotes session-minute fixed effects that flexibly absorb all time-varying content quality and session-level demand shocks, and $\delta_{ic(s)}$ are consumer-creator fixed effects. Identification therefore comes from within-session-minute variation across viewers.

Table .15 shows that, after controlling for session-minute fixed effects, individual-level liking activity has no statistically significant effect on individual gifting behavior. This result suggests that likes do not exert a direct behavioral influence on gifting decisions at the individual level once content quality is held constant. Consequently, modeling likes at the aggregate level as a proxy for content quality is appropriate for my analysis.

Table .15: Effect of Individual-level Liking on Individual-level Gifting

	gift
like	0.0003 (0.0002)
Observations	13,804,014
Session-minute FE	YES
Viewer-creator FE	YES

Note: I examine the effect of individual liking on individual gifting behavior after the content quality is controlled. I find insignificant effect of liking on gifting behavior. Standard errors are in parentheses. Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level. ¹.

C.4 Alternative Proxies for Content Quality

Moreover, I further assess the robustness of using aggregate liking activity as a proxy for content quality. In my reduced-form analysis, I use likes by others to proxy the content

quality perceived by a given consumer, thereby excluding the consumer’s own engagement. In contrast, in the structural model I use total likes in the session as the proxy for content quality. This modeling choice is driven by tractability considerations: using likes by others for each consumer would substantially increase the dimensionality of the state space and complicate the structural estimation.

Table .16 compares the estimated effects of content quality on consumers’ gifting behavior under these two alternative measures. The results are quantitatively and qualitatively similar when using total likes in the session versus likes by others. This similarity indicates that my estimates are not sensitive to the precise construction of the content-quality proxy and supports the robustness of my modeling approach in both the reduced-form and structural analyses.

Table .16: Content Quality Proxy: Others’ Likes V.S. Total Likes

	gift	
others’ likes	0.0012*** (0.0001)	
total likes		0.0012*** (0.0001)
audience size	0.0011*** (0.0004)	0.0011** (0.0004)
t	0.0003* (0.0001)	0.0003* (0.0001)
Observations	13,591,314	13,591,314
Session-minute FE	YES	YES
Control	YES	YES

Note: Standard errors are in parentheses. Significance: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors are clustered at the session level. ¹.

I do not include session-by-minute fixed effects to control for content quality for two reasons. First, in my reduced-form analysis of the competitive motive, I exploit re-entry events using a regression discontinuity-style design that compares the gifting behavior of viewers who are directly bumped in the leaderboard. At each re-entry event, only one non-focal viewer is mechanically affected. Including session-by-minute fixed effects would therefore absorb all identifying variation at the event level, rendering the estimation infeasible.

Second, session-by-minute fixed effects are not suitable for the structural analysis. Incorporating such high-dimensional fixed effects would substantially increase model complexity and obscure the economic interpretation of the estimated utility components, without improving identification of the underlying preferences. For these reasons, we rely on observed proxies for content quality and flexible time controls rather than session-by-minute fixed effects.

C.5 Interaction between Appreciative Motive and Competitive Motive

In the model, I assume that there is no interaction between the two types of motive. I conduct a heterogeneous analysis to interact the content popularity with the post variable to investigate this assumption. The specification of the analysis is as follows:

$$\begin{aligned} \text{gift}_{ist} = & \sum_{k=1}^3 \alpha_{k,k+1,like} * I\{\text{rank}_{is,t_0-1} = k, t \geq t_0\} * (\text{others' like}_{ist} - \text{mean others' like}) \\ & + \sum_{k=1}^3 \alpha_{k,k+1,aud} * I\{\text{rank}_{is,t_0-1} = k, t \geq t_0\} * (\text{audience size}_{st} - \text{mean audience size}) \\ & + \sum_{k=1}^3 \alpha_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k, t \geq t_0\} + \sum_{k=1}^2 \beta_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k\} \\ & + \lambda * \text{others' like}_{ist} + \psi * \text{audience size}_{st} + \phi * t + \theta_s + \delta_{ic(s)} + \epsilon_{ist} \end{aligned}$$

I use total likes by others to proxy content popularity while controlling for the number of viewers in the interaction term. In Table .17, I find that total likes by others do not moderate the magnitude of the competitive motive, while controlling for the audience size. This means the interaction between two motives is insignificant. Moreover, I find that the competitive motive is not significantly moderated by audience size, indicating that the argument in [Lu et al. \(2021\)](#)-viewer size positively affects gifting incentive by amplifying their social value does not apply to the competitive motive in my context.

Table .17: Interaction of Competitive and Appreciation Motive

	gift		
	1 to 2	2 to 3	3 to 4
$\alpha_{k,k+1}$	2.6361*** (0.0084)	1.3396 (0.1803)	2.7984*** (0.0051)
$\alpha_{k,k+1,like}$	0.7491 (0.4538)	0.1786 (0.8581)	0.5881 (0.5564)
$\alpha_{k,k+1,aud}$	1.3801 (0.1675)	-2.0832** (0.0372)	-0.8712 (0.3836)
Observations	198,996		
Session FE	YES		
Consumer FE	YES		
Control ¹	YES		

Note: Standard errors are in parentheses. Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level.¹ Including bumped ranking dummy, others likes, audience size, number of minutes since the start of the session.

C.6 Spillovers in Gifting Activities across Viewers

In principle, an individual’s gifting behavior can be influenced by other viewers’ gifting behavior during the same livestreaming session. In this analysis, I rule out two types of spillover in viewers’ gifting behavior: one is the herding effect — that is, viewers gift because others are gifting —after controlling for the competitive, appreciative motive and contemporaneous content quality. The other is the free-riding effect — that is, viewers refrain from giving gifts because others are contributing, after controlling for the competitive, appreciation motive, and contemporaneous content quality. Free-riding effect may happen because the gifting behavior of viewers are publicly observable. When a viewer observes others contributing a lot, she may refrain from giving more because she feels the content creator has already been paid for their hard work. Both types of gifting motives capture how others’ gifting behavior affects a person’s gifting behavior. I run the following regression to find if other viewers’ gifting amount affect a viewer’s subsequent gifting behavior

$$\text{gift}_{ist} = \alpha * \text{others' gift}_{ist-1} + \lambda * \text{others' like}_{st} + \psi * \text{audience size}_{st} + \phi * t + FE_{s,5min} + FE_i$$

In this specification, a significant coefficient of total gifts from others indicate a spillover from other viewers’ gifting behavior on ones gifting behavior, i.e., positive(negative) spillover indicates herding(free-riding) effect

Table .18: No Spillover of Gifting Activities Across Viewers

	gift
lag others’ gift	-0.0004 (0.0004)
others’ like	0.0013*** (0.00026)
Observations	1,241,826
Session-5min FE	YES
Consumer FE	YES
Controls ¹	YES

Note: Standard errors are in parentheses. Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level. ¹ Including viewer size, number of minutes since the start of the session.

In Table .18, the coefficient of lag total gift by others is insignificant, indicating that the gifting behavior of other viewers has no spillover effect on a viewer’s gifting behavior. Therefore, I rule out herding motive and free-riding motive in gifting.

C.7 End-of-Session Effects in Gifting Behavior

In this subsection, I want to rule out one potential gifting motive: that viewers gift heavily at the end of the session to become the session’s highest contributor. In the model, I assume

that viewers' competition is to obtain higher leaderboard rankings during the session. This assumption is motivated by the fact that TikTok rewards viewers by granting badges to the top three contributors, which can improve their social recognition and interaction with the content creator. I assume that viewers value the process of holding the badges.

However, another potential competition mechanism which is not captured by the model but could exist is that viewers want to become the biggest contributor of the entire session for the prominence of the session end. Due to the prominence of session ending, it is possible that being the largest contributor near the end of session have a psychological rewarding feeling on the viewers. I run the following regression to examine the end effect in gifting behavior:

$$\text{gift}_{ist} = \sum_{h=0}^{10} \alpha_h * I(t = \text{session length} - h) + \lambda * \text{others' like}_{st} + \psi * \text{audience size}_{st} + \phi * t + \delta_{ic(s)} + \epsilon_{ist}$$

In the regression, I examine how consumers change their gifting behavior in the last 10 minutes of the session. I want to see if they will significantly increase their gifting amount while they are in the session. Table .21 finds that consumers' end effect in gifting behavior is almost negligible: their gifting amount only temporarily spikes in three to four minutes to the end of the session, which is a very short span of time given the average session lasts for about , and the gifting amount is increased by 30% from the baseline, i.e., a consumer's mean gifting amount in a minute is about 12 coins. Moreover, consumers significantly decrease their gifting amount when the session ends. Overall, I find that there is no substantial end effect in consumers' gifting behavior.

C.8 Gifting and Incentives for Content Production

Besides the appreciation and competitive motives, consumers may also gift to signal that they value the content and wish the creator to continue producing similar content. If gifting serves as such a signal, it may influence subsequent content creation during the livestream. I examine this possibility by testing whether gifting affects content production along two dimensions: content quality and session continuation. I proxy content quality by the total number of likes received in a given minute and measure session continuation by whether the livestream ends in the subsequent minute.

Under the signaling hypothesis, higher gifting in the previous minute should lead to higher content quality in the next minute, as creators respond by producing content that aligns with viewers' preferences. Similarly, higher gifting should reduce the likelihood that the session ends, reflecting creators' incentives to continue the livestream when viewer support is strong.

Table .19: Negligible End Effect on Gifting Behavior

	gift
I(t=session duration)	-4.4398*** (1.3725)
I(t=session duration-1)	-0.2738 (1.1515)
I(t=session duration-2)	-0.2606 (1.2822)
I(t=session duration-3)	3.7153** (1.6501)
I(t=session duration-4)	3.9213** (1.5490)
I(t=session duration-5)	2.7275 (1.7039)
I(t=session duration-6)	0.9002 (1.2344)
I(t=session duration-7)	1.8541 (1.2466)
I(t=session duration-8)	1.4792 (1.2046)
I(t=session duration-9)	1.7977 (1.1497)
I(t=session duration-10)	1.0086 (1.1568)
Observations	31,705,888
Controls ¹	YES
Session FE	YES
Consumer $\times$ Creator FE	YES

Note: Standard errors are in parentheses. Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level. ¹ Including total like by others, audience size, number of minutes since the start of the session.

To test these predictions, I estimate the following regressions:

$$\text{attendance dummy}_{is+1} = \alpha_1 * \text{total gifts}_{is} + \lambda_1 * \text{total likes}_{is} + \gamma_1 * \text{total comments}_{is} \\ + FE_s + FE_{ic(s)}$$

$$\text{total gifts}_{is+1} = \alpha_2 * \text{total gifts}_{is} + \lambda_2 * \text{total likes}_{is} + \gamma_2 * \text{total comments}_{is} + FE_s + FE_{ic(s)}$$

Table .20 shows that gifting activity does not significantly affect subsequent content quality or session continuation. These results suggest that, within livestreaming sessions, gifting has limited influence on creators' real-time content production decisions, providing little support for a signaling motive through which viewers shape content by gifting.

Table .20: Effect of Gifting on Content Production

	total likes	session end
lag total gifts	-0.0002 (0.0002)	9.16e-8 (6.09e-8)
lag total comments	0.5515*** (0.0678)	7.53e-5*** (1.01e-05)
lag total likes	0.7418*** (0.0025)	-5.56e-06*** (2.88e-07)
Observations	1,236,889	1,236,889
Session FE	YES	YES
Controls ¹	YES	YES

Note: Standard errors are in parentheses. Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level. ¹ Including audience size, number of minutes since the start of the session.

C.9 Additional Evidence on the Instantaneous Utility of Gifting

In the model, the appreciation is modeled as an instantaneous utility, where the utility is positively related to the instantaneous content quality level. However, intuitively, people want to show appreciation for the span of content watched.

If appreciation is instantaneous, then (conditional on controls) gifts should load mainly on current quality others' like_{i,s,t}. If appreciation reflects the span of content watched, gifts should also load on lagged quality others' like_{i,s,t-1}, others' like_{i,s,t-2},...or an accumulated index. Based on this logic, I want to revisit the regression of the appreciation motive and add more lagged quality variables:

$$\text{gift}_{ist} = \sum_{\ell=0}^L \beta_{\ell} * \text{others' like}_{s,t-\ell} + \psi * \text{audience size}_{st} + \delta_{ic(s)} + \theta_s + \epsilon_{ist}$$

where I examine the lags up to 8 minutes, $L = 8$.

Table .21 presents the distributed-lag estimates of the effect of content quality on gifting behavior. The results show that gifting responds most strongly to contemporaneous content quality, with a positive and statistically significant coefficient on others’ like_{*s,t*}. The immediate lag, others’ like_{*s,t-1*}, also exhibits a positive effect. Beyond the first lag, the coefficients on lagged quality decline sharply in magnitude and become economically small, statistically insignificant, or even negative. In particular, quality measures lagged by two minutes or more have negligible explanatory power for current gifting behavior.

Taken together, these patterns suggest that viewers’ appreciation responses are highly transitory and driven primarily by real-time content quality rather than by accumulated exposure over a longer viewing window. While viewers may derive enjoyment from the span of content watched, their gifting decisions appear to reflect contemporaneous assessments of content quality rather than a stock of past enjoyment. This evidence supports the modeling assumption that the utility from showing appreciation enters as an instantaneous flow utility that depends on current content quality, and suggests that incorporating longer-lived appreciation stocks would add limited explanatory power in this setting.

C.10 Exit and Cross-Session Substitution in Competition

Our structural model focuses on competition and stagnation within a single livestreaming session and abstracts from substitution across sessions. This simplification is innocuous for evaluating within-session incentives, but it raises an important question from the platform’s perspective. If viewers who fall behind on the leaderboard simply exit the session and reallocate their gifting activity to other concurrent sessions, then stagnation may primarily reallocate revenue across creators rather than reduce total platform revenue. In that case, stagnation would be less harmful at the platform level, even if it weakens incentives and revenue within individual sessions.

To study whether leaderboard lagging induces viewers to substitute across livestreaming sessions, we decompose the decision into two sequential margins: (i) whether a gifter exits the current session, and (ii) conditional on exit, whether she switches to another session. This two-step approach is motivated by the fact that switching across sessions is mechanically possible only after exiting the current session. Estimating a single reduced-form regression that conflates these margins would therefore confound the effect of competitive discouragement on exit behavior with its effect on substitution conditional on exit.

Exit Behavior In the first step, we examine whether falling behind on the leaderboard increases the likelihood that a gifter exits the current session. Using minute-level data, we estimate the following specification:

$$\text{quit}_{i,s,t} = \beta_1 * \text{lagging}_{i,s,t} + X_{i,s,t} + \delta_s + \gamma_{i,c(s)} + \varepsilon_{i,s,t},$$

Table .21: Evidence of Transitory Appreciation Response to Content Quality

	gift
others' like _{<i>i,s,t</i>}	0.0004*** (0.0001)
others' like _{<i>i,s,t-1</i>}	0.0010*** (0.0003)
others' like _{<i>i,s,t-2</i>}	0.00005 (0.0004)
others' like _{<i>i,s,t-3</i>}	-0.0004** (0.0002)
others' like _{<i>i,s,t-4</i>}	-0.0001 (0.0001)
others' like _{<i>i,s,t-5</i>}	-0.0001* (0.00007)
others' like _{<i>i,s,t-6</i>}	0.00006 (0.00007)
others' like _{<i>i,s,t-7</i>}	0.00006 (0.00006)
others' like _{<i>i,s,t-8</i>}	0.00003 (0.0001)
Observations	10,245,613
Controls ¹	YES
Session FE	YES
Viewer × Creator FE	YES

Note: Standard errors are in parentheses. Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level. ¹ Including audience size, number of minutes since the start of the session.

where $\text{quit}_{i,s,t}$ is an indicator for whether gifter i exits session s at minute t . The variable $\text{lagging}_{i,s,t}$ captures the gifter’s competitive position on the leaderboard, measured either by her ranking status or by the score distance to the third-ranked gifter. The vector $X_{i,s,t}$ includes controls for session activity (e.g., likes by other viewers) and time since the session started. Session fixed effects control for session-level characteristics, while viewer-streamer fixed effects absorb time-invariant individual heterogeneity in exit propensity.

Substitution Conditional on Exit In the second step, we restrict attention to minutes in which a gifter exits the session and examine whether competitive lagging affects the likelihood that the exit is accompanied by a switch to another session. Specifically, we estimate:

$$\text{switch}_{i,s,t} = \beta_2 * \text{lagging}_{i,s,t} + X_{i,s,t} + \delta_s + \gamma_{i,c(s)} + \varepsilon_{i,s,t},$$

where $\text{switch}_{i,s,t}$ is an indicator equal to one if gifter i exits session s at minute t and enters another session immediately (i.e., within 10 minutes) thereafter. By conditioning on exit, this specification isolates the substitution margin from the extensive margin of leaving the session altogether. Together, the two regressions allow us to distinguish whether leaderboard discouragement primarily leads to disengagement from livestreaming or to active reallocation of attention and gifting across sessions.

Table .22: Exit and Cross-Session Substitution

	quit		switch	
	(1)	(2)	(3)	(4)
Ranking status	$-1.04 \times 10^{-4}***$ (1.13×10^{-5})		1.50×10^{-6} (2.60×10^{-6})	
Distance to No. 3 score		$7.9 \times 10^{-8}**$ (2.9×10^{-8})		1.5×10^{-9} (4.6×10^{-9})
Observations	22,586,273	27,701,618	1,090,206	1,207,042
Session FE	YES	YES	YES	YES
Viewer-creator FE	YES	YES	YES	YES
Controls ¹	YES	YES	YES	YES

Note: Standard errors are in parentheses. Significance: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors are clustered at the session level. The dependent variable in Columns (1)–(2) is an indicator for exiting the current session. The dependent variable in Columns (3)–(4) is an indicator for switching to another session, conditional on exit. ¹ Including total likes by others, number of minutes since the start of the session.

Table .22 presents evidence on how leaderboard lagging affects viewers’ exit and substitution behavior. Columns (1) and (2) show that falling behind on the leaderboard significantly increases the likelihood that a gifter exits the current session: lower ranking status and larger score gaps relative to the third-ranked gifter are both associated with a higher probability of exit. This pattern is consistent with competitive discouragement within sessions.

In contrast, Columns (3) and (4) show that conditional on exit, leaderboard position has

no statistically significant effect on the likelihood that a gifter switches to another session. Neither ranking status nor score distance predicts substitution behavior once the decision to exit has been made. Together, these results suggest that competitive stagnation primarily induces disengagement from the focal session rather than active reallocation of gifting activity to other sessions. As a result, stagnation is unlikely to be merely redistributing revenue across creators, but instead reflects a reduction in overall viewer engagement at the platform level.

A potential concern with our substitution analysis is that we only observe switching among creators included in our sample, which consists of creators featured on the TikTok LIVE front page. As a result, we do not capture substitution toward less visible creators outside the sample. We argue that this limitation is unlikely to materially affect our conclusions. Given the highly skewed distribution of attention on livestreaming platforms, viewers who substitute across sessions are most likely to reallocate their attention toward highly visible and popular creators. Our sample, therefore, captures the most relevant margin of cross-session substitution. To the extent that unobserved switching occurs toward less prominent sessions, our estimates should be interpreted as an upper bound on substitution toward competing high-visibility creators.

D Additional Analyses for Counterfactual Design

D.1 Full Simulation Results for Discounted Score Design

Table .23 shows how total gifts, turnover rate and number of gifters changes with discount factor.

D.2 Content Production under Alternative Scoring Rules

Regarding the counterfactual about score design, I want to prove that shuffling the ranking among gifters will not lead to changes in content production, i.e., content quality and session duration. To validate such assumption, I run the following regressions at the session-minute level:

$$\begin{aligned} \text{total likes}_{st} &= \alpha_1 * \text{total gifts}_{st-1} + \beta_1 * \text{total comments}_{st-1} \\ &+ \gamma_1 * \text{total likes}_{st-1} + \psi_1 * \text{audience size}_{st} + \phi_1 * \text{since start}_{st} + FE_s \end{aligned}$$

$$\begin{aligned} \text{session end}_{st} &= \alpha_2 * \text{total gifts}_{st-1} + \beta_2 * \text{total comments}_{st-1} \\ &+ \lambda_2 * \text{total likes}_{st-1} + \psi_2 * \text{audience size}_{st} + \phi_2 * \text{since start}_{st} + FE_s \end{aligned}$$

I investigate content production during the livestreaming session. I consider two aspects of content production: one is about content quality, proxied by the total likes in the session. The other is about the session length decision, measured by the probability of the session

Table .23: Full Counterfactual Outcomes under Alternative Score Discounting

β	total gifts (\$)	% change in revenue	turnover rate	number of gifters
1.00	39.16	0.0%	0.0541	7.10
0.95	39.16	0.0%	0.0541	7.10
0.90	39.16	0.0%	0.0541	7.10
0.85	39.16	0.0%	0.0541	7.10
0.80	39.16	0.0%	0.0541	7.10
0.75	39.95	2.0%	0.0581	7.10
0.70	43.65	11.5%	0.0863	7.17
0.65	46.71	19.3%	0.1290	7.49
0.60	46.79	19.5%	0.1288	7.50
0.55	46.87	19.7%	0.1291	7.50
0.50	46.67	19.2%	0.1295	7.49
0.45	46.40	18.5%	0.1566	7.73
0.40	46.26	18.1%	0.1566	7.73
0.35	46.23	18.1%	0.1568	7.74
0.30	46.09	17.7%	0.1590	7.74
0.25	45.79	16.9%	0.1586	7.75
0.20	45.77	16.9%	0.1587	7.76
0.15	44.18	12.8%	0.1665	7.77
0.10	43.98	12.3%	0.1666	7.78
0.05	43.85	12.0%	0.1673	7.78

Table .24: Effect of Viewer Engagement on Content Production

	total likes	session end
lag total gifts	-0.0002 (0.0002)	9.16e-8 (6.09e-8)
lag total comments	0.5515*** (0.0678)	7.53e-5*** (1.01e-05)
lag total likes	0.7418*** (0.0025)	-5.56e-06*** (2.88e-07)
Observations	1,236,889	1,236,889
Session FE	YES	YES
Controls ¹	YES	YES

Note: Standard errors are in parentheses. Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level. ¹ Including audience size, number of minutes since the start of the session.

ending at each minute. In Table .24, past comments and past likes will affect content production, in terms of subsequent content quality and the probability of session ending.

Therefore, I run the following regression to investigate whether consumers change their commenting and liking behavior when their ranking is bumped down on the leaderboard during re-entry events.

$$\begin{aligned} \text{comment}_{ist} &= \sum_{k=1}^4 \alpha_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k, t \geq t_0\} + \sum_{k=1}^3 \beta_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k\} \\ &\quad + \lambda * \text{others' like}_{ist} + \psi * \text{audience size}_{st} + \phi * t + \theta_s + \delta_{ic(s)} + \epsilon_{ist} \\ \text{like}_{ist} &= \sum_{k=1}^4 \alpha_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k, t \geq t_0\} + \sum_{k=1}^3 \beta_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k\} \\ &\quad + \lambda * \text{others' like}_{ist} + \psi * \text{audience size}_{st} + \phi * t + \theta_s + \delta_{ic(s)} + \epsilon_{ist} \end{aligned}$$

I find that viewers generally do not significantly change their commenting and liking behavior after being demoted on the leaderboard. One except is that No.1 gifter reduces her comment activity. However, the magnitude is so small that it has a negligible effect on content quality; i.e., total likes drop by $0.55 * 0.01 = 0.0055$. As a result, I can conclude that viewers will not change their commenting and liking behavior after being demoted on the leaderboard when their rankings on the leaderboard get shuffled; therefore, the ranking shuffle will not change the content production process through change in commenting and liking activities. Moreover, Table .24 suggests that the gifting amount has an insignificant effect on content quality and the probability of session end. Even if in the counterfactual that gifts under discounted score increases, it does not affect the content production process through boosting content creators' emotions or ability to produce high-quality content. This is consistent with the fact that my sample only includes content creators who were recommended on the TikTok front page, meaning they are popular and mature in the content production process. Therefore, changes in gifting amount do not affect their content production.

D.3 Cross-Session Revenue Correlation

In my counterfactual analysis, I optimally design the leaderboard to maximize the gifting revenue within sessions. However, maximizing within-session gifting revenue may not be optimal if increasing gifting in one session would crowd out the gifting contribution in future sessions. Some of the reasons that might happen are: (1) Consumers have a budget constraint on how much they can contribute. If they contributed a lot in one session, it may deplete her budget for a while. Therefore, she cuts back her spending in future sessions; (2) Consumers regret spending too much money on the creator. Hence, after spending a large amount of cash on the creator, the consumer stops visiting the channel or visiting the session, but stops contributing for a while; (3). Consumers contributed a lot in past sessions do not need to contribute in the current session and can still get attention from the content creator because

Table .25: Commenting and Liking for Bumped Viewers

	comment	like
$I\{\text{rank}_{is,t_0-1} = 1, t \geq t_0\}$	-0.0133** (0.0066)	-0.4817 (0.4668)
$I\{\text{rank}_{is,t_0-1} = 2, t \geq t_0\}$	-0.0072 (0.0059)	-0.3183 (0.4291)
$I\{\text{rank}_{is,t_0-1} = 3, t \geq t_0\}$	0.0022 (0.0070)	-0.1359 (0.4751)
$I\{\text{rank}_{is,t_0-1} = 4, t \geq t_0\}$	0.0071 (0.0069)	0.4853 (0.5181)
others' like	-0.000009 (0.000006)	0.0057*** (0.0005)
Observations	198,996	198,996
Session FE	YES	YES
Viewer $\times$ Creator FE	YES	YES
Controls ¹	YES	YES

Note: Standard errors are in parentheses. Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level. ¹ Including audience size, number of minutes since the start of the session, and rank dummy.

the content creators can still recognize the name of frequent contributors. In the following regressions, I investigate how a viewer's gifting amount within a session affects her subsequent session's attendance as well as gifting amount, while controlling for her engagement within the session, i.e., number of likes and number of comments sent, session fixed effect and creator-viewer fixed effect.

$$\text{attendance dummy}_{is+1} = \alpha_1 * \text{total gifts}_{is} + \lambda_1 * \text{total likes}_{is} + \beta_1 * \text{total comments}_{is} \\ + FE_s + FE_{ic(s)}$$

$$\text{total gifts}_{is+1} = \alpha_2 * \text{total gifts}_{is} + \lambda_2 * \text{total likes}_{is} + \beta_2 * \text{total comments}_{is} + FE_s + FE_{ic(s)}$$

where s is the s th session of the creator if I order all sessions by their broadcasting time.

I find that a viewer's gifting in sessions s has a positive effect on her next session's gifting amount and attendance probability while I control for how much she likes the content through total likes and commenting activities. Meaning my counterfactual analysis on the optimal design of the scoring systems is a conservative estimate of the revenue improvement, because it does not account for the positive correlation in revenue across sessions. Moreover, my estimation about the contribution of the leaderboard is also conservative, given that I did not fully capture that leaderboard boost revenue can positively drive inter-session dependence.

Table .26: Intertemporal Effect of Gifting

	attendance dummy	total gifts
lag total gift	1.81e-06*** (3.89e-07)	0.07*** (0.02)
lag total comment	8.83e-04*** (1.12e-04)	0.18 (0.14)
lag total like	2.40e-05*** (1.69e-06)	-0.001 (0.001)
Observations	4,850,541	4,850,541
Session FE	YES	YES
Creator-consumer FE	YES	YES

Note: Standard errors are in parentheses. Significance: *p<0.1; **p<0.05; ***p<0.01.

D.4 Discounted Scoring and Preferences for Ranking

Regarding the counterfactual about discount score design, an assumption of the analysis is that viewers' valuation of the badges, i.e., the rank-based utility, does not change due to the discounted score. The following are some potential reasons that could lead to a change in rank-based utility due to discounted scores:

- 1 Viewers exhibit greater respect for the all-time biggest contributor than the biggest contributor for the last 30 minutes. Since discounted scores reflect more recent contributions of viewers, the social recognition from the same badges might decrease due to discounted scores. Because achieving all-time contributors (top 3 positions when there is no discount) is more respectful than achieving past-30-minute contributors.
- 2 Viewers exhibit greater respect for the top contributors under the cumulative gifting score than those under the discounted score, given that viewers generally need to contribute less under discounted scores in order to achieve the top ranking.
- 3 Viewers exhibit greater respect for the top contributors when the competition is more intense. Since the competition becomes more intense when the discounted scores are implemented, given all gifters keep up to their game at all times, viewers might show greater respect to those top contributors who win when discounted scores are implemented and the competition is more intense.
- 4 Identities of top contributors change more frequently, and fulfilling the top contributors' requests becomes more difficult. Given that another value of badge comes from improving the interaction with content creators through the comment section, I should expect fulfilling the top contributors' requests to become more difficult when the identities of the top contributors are shuffled more frequently.

Given that the discounted scoring system has never implemented on TikTok, I am not able to draw a conclusive statement on whether the above conjectures stand true. However, I am able to test those mechanisms indirectly through the heterogeneous analysis in the re-entry event. The underlying logic behind the heterogeneous analysis is that given re-entry timing is plausibly exogenous, there is substantial variation in the session environment at the time of re-entry. As a result, I am able to test viewers' competitive motive, i.e., their contemporary attitudes towards the ranking, across a wide range of competitive environments.

Case 1: For the first point, I conduct a heterogeneous analysis of the re-entry event to see how viewers' preferences for leaderboard positions change depending on whether the re-entry occurs within the first 30 minutes of the session. Because if re-entry events happen early on, viewers may respect the top contributors less because the social recognition is based on a shorter time span of gifting contribution. Versus, if the re-entry happens beyond the first 30 minutes of the session, viewers may respect top contributors more if the score is based on total contribution in a longer time span. If that is true, I should observe a significant difference in gifting response in re-entry event depending on whether the re-entry timing is within 30 minutes or not. To test this, I run the following regression:

$$\begin{aligned} \text{gift}_{ist} = & \sum_{k=1}^3 \alpha_{k,k+1,t_0 \leq 30} * I\{\text{rank}_{is,t_0-1} = k, t \geq t_0, t_0 \leq 30\} \\ & + \sum_{k=1}^3 \beta_{k,k+1,t_0 \leq 30} * I\{\text{rank}_{is,t_0-1} = k, t_0 \leq 30\} \\ & + \sum_{k=1}^3 \alpha_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k, t \geq t_0\} + \sum_{k=1}^2 \beta_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k\} \\ & + \lambda * \text{others' like}_{ist} + \psi * \text{audience size}_{st} + \phi * t + \theta_s + \delta_{ic(s)} + \epsilon_{ist} \end{aligned}$$

Table .27: Heterogeneous Analysis for Re-entry Event Happening within the First 30 Minutes

	Gift by bumped-down viewers		
	1 to 2	2 to 3	3 to 4
$\alpha_{k,k+1,t_0 \leq 30}$	9.7942 (12.4428)	-6.1446 (5.4137)	15.4794 (16.1854)
Observation	198,996		
Controls	YES		
Session FE	YES		
Viewer $\times$ Creator FE	YES		

Note: Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level.

Table .27 finds that bumped viewers' change in gifting behavior does not significantly vary across re-entry timing, within 30-minute V.S. not within 30-minute. This result indicates that viewers' competitive motive, i.e., preference for rankings, does not depend on whether

the score is evaluated based on total contribution over the last 30 minutes or over a longer period.

Case 2: For the second point, I conduct the heterogeneous analysis in the re-entry event to see how viewers' preferences for the positions on the leaderboards change depending on their cumulative gifting amount(score) at the moment of re-entry. I run the following regression:

$$\begin{aligned}
\text{gift}_{ist} = & \sum_{k=1}^3 \alpha_{k,k+1,\text{score}} * I\{\text{rank}_{is,t_0-1} = k, t \geq t_0\} * \text{score}_{s,t_0-1} \\
& + \sum_{k=1}^3 \beta_{k,k+1,\text{score}} * I\{\text{rank}_{is,t_0-1} = k\} * \text{score}_{s,t_0-1} \\
& + \sum_{k=1}^3 \alpha_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k, t \geq t_0\} + \sum_{k=1}^2 \beta_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k\} \\
& + \lambda * \text{others' like}_{ist} + \psi * \text{number of viewers}_{st} + \phi * t + \theta_s + \delta_{ic(s)} + \epsilon_{ist}
\end{aligned}$$

Table .28: Heterogeneous Analysis for Re-entry Event Depending on the Score of Bumped Viewers

	Gift by bumped-down viewer		
	1 to 2	2 to 3	3 to 4
$\alpha_{k,k+1,\text{score}}$	0.0087 (0.0070)	0.0135 (0.0187)	-0.0025 (0.0025)
Observation	198,996		
Controls	YES		
Session FE	YES		
Viewer $\times$ Creator FE	YES		

Note: Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level.

Table .28 finds that viewers' gifting response to the re-entry event insignificantly depend on the bumped viewers' score at the moment of re-entry. This indirectly suggests that viewers' utility from the ranks does not depend on how much score the viewers have accumulated.

Case 3: For the third point, I conduct the heterogeneous analysis in the re-entry event to see how viewers' preferences for the positions on the leaderboards change depending on the competition level at the time of re-entry, which is measured by the HHI of gifters' cumulative

gifting contribution. I run the following regression:

$$\begin{aligned}
\text{gift}_{ist} = & \sum_{k=1}^3 \alpha_{k,k+1,\text{HHI}} * I\{\text{rank}_{is,t_0-1} = k, t \geq t_0\} * \text{HHI}_{s,t_0-1} \\
& + \sum_{k=1}^3 \beta_{k,k+1,\text{HHI}} * I\{\text{rank}_{is,t_0-1} = k\} * \text{HHI}_{s,t_0-1} \\
& + \sum_{k=1}^3 \alpha_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k, t \geq t_0\} + \sum_{k=1}^2 \beta_{k,k+1} * I\{\text{rank}_{is,t_0-1} = k\} \\
& + \lambda * \text{others' like}_{ist} + \psi * \text{number of viewers}_{st} + \phi * t + \theta_s + \delta_{ic(s)} + \epsilon_{ist}
\end{aligned}$$

Table .29: Heterogeneous Analysis for Re-entry Event Depending on the HHI

	Gift by bumped-down viewer		
	1 to 2	2 to 3	3 to 4
$\alpha_{k,k+1,\text{HHI}}$	-6.3463 (8.9702)	-28.3624 (28.5005)	4.1274 (5.5676)
Observation	198,996		
Controls	YES		
Session FE	YES		
Viewer $\times$ Creator FE	YES		

Note: Significance: *p<0.1; **p<0.05; ***p<0.01. Standard errors are clustered at the session level.

Table .29 finds that viewers' gifting response to the re-entry event does not significantly depend on the competition intensity at the moment of re-entry. This indirectly suggests that viewers' preference for the ranks does not depend on whether the competition intensity at the moment of re-entry.

For the forth point, I find the turnover rate of top 3 gifters is 0.073 under the optimal score discount in Table 9, which is still low. Therefore, I expect the shuffle in gifter identity does not impose great difficulty in the content production process to fulfill the top contributors' requests.

D.5 Behavioral Responses to Discounted Scoring

Regarding the counterfactual analysis of discounted scoring, I assume that viewers do not experience substantial frustration in response to score discounting that would lead to a behavioral backfire. While I do not directly observe frustration, several considerations suggest that this assumption is reasonable in the context I study.

First, effective communication of the discounting rule can mitigate negative reactions. If viewers clearly understand that scores decay over time according to a known rule, they can adjust their gifting strategies accordingly rather than perceiving discounts as unexpected

losses. Figure 6 plots the gifting dynamics for the No.1 gifter in the entire session. Consistent with this intuition, it shows that under discounted scoring, gifting is more evenly distributed over the session: top contributors reduce spending early and increase spending later. This smoothing pattern makes it unlikely that viewers concentrate spending early in the session only to experience large subsequent discounts, a scenario that would be most likely to generate frustration.

Second, discounted scoring can be framed as a mechanism that enhances fairness and participation by giving viewers opportunities to compete for top positions regardless of when they enter the session. My counterfactual results in Table .23 show that discounted scoring encourages participation by improving the likelihood of achieving high leaderboard ranks, which can offset potential frustration from score decay.

Finally, in my counterfactual design, score discounting occurs at a relatively low frequency—once every 30 minutes—which further limits the salience of score decay and reduces the likelihood of persistent or repeated frustration. Taken together, these considerations indicate that discounted scoring is unlikely to generate strong behavioral backfire effects that would undermine its revenue-enhancing potential.

D.6 Industrial Evidence on Planned Session Duration

Industry guidance for professional livestreamers emphasizes that successful sessions are typically structured and planned in advance rather than conducted spontaneously. Practitioner-oriented materials consistently highlight the importance of defining a clear run of show prior to going live, including the sequencing of content segments, planned audience interactions, and approximate time allocation across the session.

Figure .7 illustrates this practice using an industry training resource for livestreaming entrepreneurs. The video stresses pre-planning the flow and timing of the session and coordinating interactive moments in advance, treating session duration as a component of content production rather than an outcome determined endogenously during the broadcast.

These practices are especially common among established creators, for whom livestreaming is a professionalized activity aimed at maximizing engagement and gifting revenue. Because platforms and creators share gifting-based revenues, platforms can directly obtain information about planned session structure and expected duration prior to going live. Together, this evidence supports interpreting session duration for mature creators as an *ex ante* expected characteristic rather than an *ex post* realized outcome.

Figure .7: Industry Guidance on Planning Livestreaming Session Structure and Timing



E Estimation Details

E.1 Full Specification for State Transition

The state transition process $k'(k; ev)$ determines how the state vector may change from a particular value of k to k' , where ev describes an event that happens in the game. The value of ev will determine gifters' expectations for the session's future state. Now we want to take the perspective of an incumbent in the session and describe how her state vector changes in eight different events:

1. $ev = gift_f(s')$: a focal incumbent increases her score from s to s' by gifting. In this case, $k'(k; gift_f(s')) = (g, s', r', D', l, N^{idle}, h)$. D' will be the new distribution of scores when the gifter has score level s' . In particular, $D' = D - e_s + e_{s'}$, where e_j is a vector with 1 in element j and 0 in all other elements. r' will be the new rank obtained by the gifter. The new rank will be determined as follows according to the gifter's new score slot s' and the number of gifters at or above that slot. Denote the state vector $\tilde{D} = D - e_s$ as describing the states of all incumbents except the focal one, and $\tilde{d}_l$ as the each element of $\tilde{D}$. If the focal gifter spend positive amount, her rank is $r' = \min\{\sum_{s'' > s} \tilde{d}_{s''} + 1, 4\}$.
2. $ev = gift_o(s^m)$: Another incumbent m who has score s^m chooses to increase her score to $s^{m'}$. $k'(k; gift_o(s^m)) = (g, s, r', D', l, N^{idle}, h)$, where r' is the new rank of the focal gifter after incumbent m changes their score, and D' is the corresponding new distribution of scores. Similar to event $gift_f$, $D' = D - e_{s^m} + e_{s^{m'}}$. The focal gifter's rank will change as follows: if the non-focal incumbent initially has a lower ranking than the focal gifter ($r^m \geq r$), and later surpasses the focal gifter's score ($s^{m'} > s$), the

focal gifters' rank is bumped down, $r' = \min(r + 1, 4)$. Otherwise, its rank does not change, i.e., $r' = r$.

3. $ev = entry$: A new entrant enters the session. $k'(k; entry) = (g, s, r, D', l, N^{idle}, h)$, where D' is the post-entry score distribution. Note that since new entrants come in with a score of 0, $d'_0 = d_0 + 1$. The focal gifter's rank will remain unchanged.
4. $ev = exit_f$: The focal incumbent i chooses to exit the session. The transition is denoted as $k'(k; exit_f) = (g', s, r', D', l, N^{idle'}, h)$. Since the incumbent has exited, $g' = 0$. When the incumbent has exited, their rank becomes irrelevant, so I set $r' = 4$. The distribution of scores will also change, in that the entry for score s will be reduced by 1: $d'_s = d_s - 1$. Finally, since there is now another idle gifter, $N^{idle'} = N^{idle} + 1$.
5. $ev = exit_o(m)$: Another incumbent m exits the session. Suppose the incumbent m who exits has score s^m , and rank r^m , the state transition for gifter i is denoted as $k'(k; exit_o(m))$. The exit of incumbent m changes the score distribution, $d'_{s^m} = d_{s^m} - 1$, increases the number of idle gifters by one, $N^{idle'} = N^{idle} + 1$, and sometimes focal incumbent i 's ranks if the exited incumbent m originally ranked higher than focal incumbent i .
6. $ev = re - entry(s^m)$: An idle gifter, m , re-enters the session. $k'(k; re - entry(s^m)) = (g, s, r', D', l, N^{idle'}, h)$. D' is the new score distribution, where $d'_{s^m} = d_{s^m} + 1$ and s^m is the idle gifter's score at the time they exited the session. r' is the focal gifter's new rank. It will evolve in the same way as in item 2, when the event $gift_0(s^m)$ occurs. The number of idle gifters decreases, $N^{idle'} = N^{idle} - 1$.
7. $ev = quality(l')$: An exogenous change in content quality from l to l' occurs. The state transition process for content quality is shown in Equation (3). The state transition for this event will be $k'(k; quality(l')) = (g, s, r, D, l', N^{idle}, h)$
8. $ev = latent$: An exogenous change in latent variable from off ($h = 0$) to on ($h' = 1$). The state transition for this event will be $k'(k; latent) = (g, s, r, D, l, N^{idle}, 1)$

E.2 Maximum Likelihood Estimation for Entry and Re-entry

I estimate $\lambda_e(W_t^e; \theta^e) = (W_t^e)' \theta^e$ and $\lambda_{re}(W_t^e; \theta^{re}) = (W_t^e)' \theta^{re}$ using the maximum likelihood method under an exponential arrival assumption. For entry event, I assume that there is one potential entrant eligible for entry opportunity as long as the number of active gifters (incumbents) is below seven. For re-entry event, I assume that all idle gifters are eligible for re-entry opportunity as long as the number of active gifters (incumbents) is below seven.

Now consider a sample of S sessions where each session s , there is a sequence of events (entry, exit, re-entry, gift, change of likes, and end of session). The number of event is $H(s)$ in session s , with event time happening at $\{t_0^s, t_1^s, t_2^s, \dots, t_{H(s)}^s\}$ minutes of the session (measured relative to the session start). $t_0^s = 0$ is the start of the session. Consider an h th event happens at minute t_h^s , the entry rate is fixed during time interval between t_{h-1}^s and t_h^s , i.e., $\lambda_e(W_{t_h^s}; \theta^e)$, where the session characteristics between that interval is $W_{t_h^s}$. Similarly, the re-entry rate is fixed between t_{h-1}^s and t_h^s at $\lambda_{re}(W_{t_h^s}; \theta^{re})$. I respectively define the number of incumbent and idle gifters to be $N^{incumbent}(t_h^s)$ and $N^{idle}(t_h^s)$ between t_{h-1}^s and t_h^s .

I estimate the entry parameters by maximizing the log-likelihood function of entry events:

$$\log L(\theta^e) = \sum_s \sum_{h=1}^{H(s)} (N^{incumbent}(t_h^s) < N) * [(event_h = \text{entry}) * \log(\lambda_e(W_{t_h^s}; \theta^e)) - \lambda_e(W_{t_h^s}; \theta^e) * (t_h^s - t_{h-1}^s)]$$

Similarly, I estimate the re-entry parameters by maximizing the log-likelihood function of re-entry events:

$$\log L(\theta^{re}) = \sum_s \sum_{h=1}^{H(s)} (N^{incumbent}(t_h^s) < N) * [(event_h = \text{re-entry}) * \log(\lambda_{re}(W_{t_h^s}; \theta^{re}) * N^{idle}) - \lambda_{re}(W_{t_h^s}; \theta^{re}) * N^{idle} * (t_h^s - t_{h-1}^s)]$$

where $event_h$ indicate the event type: entry, exit, re-entry, like transition or session end.

E.3 Maximum Likelihood Estimation for Like Transitions

Now consider a sample of S sessions where each session s I observe a sequence of content quality changes occurring $H(s)$ times, with event time happening at $\{t_0^s, t_1^s, t_2^s, \dots, t_{H(s)}^s\}$ minutes of the session (measured relative to the session start). $t_0^s = 0$ is the start of the session. The h th change in quality during session s moves the quality from $l(h-1, s)$ to $l(h, s)$. The time interval of content quality is at $l(h-1, s)$ before transit to $l(h, s)$ is $t_h^s - t_{h-1}^s$. I estimate all combinations of $q_{ll'}$ by using maximum likelihood estimator:

$$\hat{Q}_q = \arg \max_{Q_q} \sum_{s \in S} \left\{ \sum_{h=1}^{H(s)} \left[\log(q_{l(h-1,s)l(h,s)}) - \sum_{l' \neq l(h-1,s)} q_{l(h-1,s)l'} \cdot (t_h^s - t_{h-1}^s) \right] - \sum_{l' \neq l(H(s),s)} q_{l(H(s),s)l'} \cdot (T_s - t_{H(s)}^s) \right\}$$

where

$$\hat{q}_{l,l'} = \frac{\sum_{s \in S} \sum_{h=1}^{H(s)} I(l(h-1, s) = l, l(h, s) = l')}{\sum_{s \in S} \left[\sum_{h=1}^{H(s)} I(l(h-1, s) = l) \cdot (t_h^s - t_{h-1}^s) + I(l(H(s), s) = l) \cdot (T_s - t_{H(s)}^s) \right]}$$

E.4 Estimates for Decision Chance, Entry, Re-entry, Like Transitions, and Session Termination

Table .30: Estimates for Decision Chance, Entry, Re-entry, Like Transitions, and Session Termination

Gift&Exit	0.15 (0.07)
Entry	
weekdays	0.05 (0.24)
weekend	0.06 (0.38)
high quality	0.04 (0.32)
Re-entry	
weekdays	0.004 (0.04)
weekend	0.003 (0.04)
high quality	0.001 (0.04)
Quality	
low to high	0.20 (0.006)
high to low	0.10 (0.003)
Latent state transition	0.008 (0.003)
Session end	0.10 (0.04)